MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, READ ONLY, PROGRAMMABLE, DIGITAL, KEY CODE GYF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic programmable read only memory module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

*2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Application requirements. This specification establishes the general requirements and test procedures for a programmable read only memory (PROM) module. When a SEMP user determines a requirement for a PROM module, the user shall prepare a design approval request (DAR) and submit the DAR to the SEMP Design Review Activity (SEMP-DRA) in accordance with MIL-STD-1378. The module user shall prepare a separate detailed module control document.

3.2.1 Design approval request (DAR) requirement. When a user determines a requirement for a GYF module, the user shall prepare a DAR and submit the DAR to the SEMP-DRA in accordance with MIL-STD-1378. The purpose of the DAR requirement is to notify the SEMP-DRA of additional GYF module users and enable the SEMP-DRA to assign GYF dash numbers in accordance with 3.2.2.1. The user shall include in the DAR the estimated number of GYF dash numbers required and the user's detailed module document number.

3.2.2 Detailed module document. The GYF module user shall prepare a separate detailed module document. This document shall carry a system drawing number and all system parts lists and the module shall reference the user's detailed module document number and not M28787/169. The detailed module document shall specify, as a minimum, the module key code as specified in 3.2.2.1, detailed module document number marking, firmware pattern (truth table), module programming technique (method) and marking as specified in 3.2.2.3, detailed electrical test requirements for a programmed module, and MIL-M-28787/169 requirements.

3.2.2.1 Module key code assignment. The module user is free to select a marking scheme but the module marking scheme (key code) shall not be a three letter key code other than GYF or a letter-number-letter key code which has been reserved for read only memory (ROM) modules by the SEMP-DRA. The SEMP-DRA has developed a controlled marking scheme which utilizes the GYF key code. Upon request by a DAR form the SEMP-DRA will assign a block of dash numbers which the user may mark on the module fin to distinguish between the different GYF programmed modules. This block of dash numbers will not be issued to other users even if all the dash numbers assigned are not used. For example, company A estimates a need for 75 unique GYF modules. Company A completes a DAR and requests a block of 75 GYF-dash numbers. The SEMP-DRA would assign GYF-01200 through GYF-01275 to company A. This assignment indicates the SEMP-DRA had already assigned GYF-00001 through GYF-01199 to other users. Company A would then mark GYF-01200 to the first pattern requirement and continue until GYF-01275 was used. If additional dash numbers are required, the user shall request them from the SEMP-DRA. NOTE: Under no circumstances shall a user assign a key code or dash numbers.

3.2.2.2 Firmware pattern. The module firmware pattern (truth table) shall be specified in the detailed module document. If the key code-dash number scheme of 3.2.2.1 is used, the different firmware patterns are specified by dash numbers. The different firmware patterns could be specified as a part number to the detailed module document. For example, if the detailed document number was 2658555, the first firmware pattern would be specified as 2658555P1. The nth firmware pattern would be specified as 2658555Pn.

3.2.2.3 Program technique and marking. The detailed programming technique (method) shall be specified in the detailed module document. Because of the numerous unique programming methods, the user must determine the number of unique programming methods allowed. The detailed module document shall specify the allowed programming procedure and the corresponding module marking procedure. As a minimum, the detailed module document shall specify: The exact step-by-step programming procedure for each programming option, minimum and maximum programming electrical parameters, maximum number of programming attempts to program a selected bit, programming time and duty cycle, maximum module temperature, and the exact module voltages and output load current to perform program verifications. For example, company A has surveyed the potential module vendors and programming techniques and has elected to allow two programming methods. Because of potential programming technique changes, company A has decided to specify each programming method by two separate programming documents. The detailed module document specifies that the module shall be programmed in accordance with the selected program method and the module shall be marked with the corresponding program document number including revision level as applicable.

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3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787. All unprogrammed modules shall be tested on a correlated test systems. All programmed modules shall be tested as specified in the user's detailed module document.

3.4 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.711 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I, the user's detailed module document (see 3.2.2), and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389, the user's detailed module document (see 3.2.2), and as specified herein.

3.6.1 Key code. The module key code shall be as specified in the user's detailed module document.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and in the user's detailed module document, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.7 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.8 Unprogrammed GYF modules. Unprogrammed modules shall be as specified in the user's detailed module document and as specified herein.

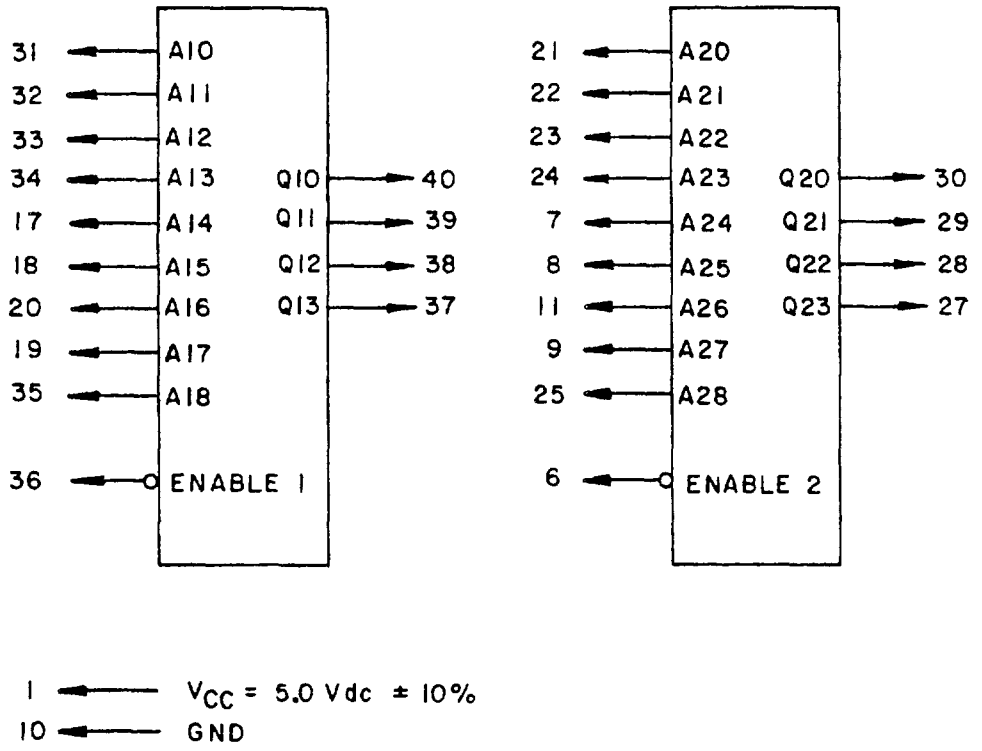
3.8.1 Unprogrammed output state. All unprogrammed module outputs shall be in the same output state (high or low level) and the state shall be as specified in the user's detailed module document.

3.8.2 Module programming. All modules shall be capable of being programmed using the module connector pins and using the appropriate programming technique as specified in the user's detailed module document.

3.8.3 Module programming technique marking. The programming technique necessary to program an individual module shall be marked on that individual module by the module vendor in accordance with the user's detailed module document.

3.9 Freeze-out/nichrome etch test. The freeze-out or nichrome etch test shall be performed on all PROM integrated circuits if the fuse is formed with a thin film metallic (including nichrome) material. The test freeze-out shall be performed in accordance with 4.6. The nichrome etch test shall be performed in accordance with 4.7. The freeze-out or nichrome etch test shall be performed on all devices selected for group B testing of method 5005 for class B of MIL-STD-883 and one failure shall constitute failure of the lot. Deviations to the freeze-out/nichrome etch test shall be specified in the detailed module document.

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NOTE: Unused pins: 2, 3, 4, 5, 12, 13, 14, 15, 16, and 26.

FIGURE 1. GYF module function diagram, programmable read only memory.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation, frame to module pins	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
A2-A11	Isolation, unused to used pins	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B18	Input breakdown current, address, V_{IH} = 5.5 V	III	S	0	1	0	1	0	1	0	1	mA
B19,B20	Input breakdown current, enable, V_{IH} = 4.5 V	III	S	-0.1	1.0	-0.1	1.0	-0.1	1.0	-0.1	1.0	mA
C1-C8	Output short circuit current	IV	S	-15	-90	-15	-90	-15	-90	-15	-90	mA
D1-D8	V_{OUT} (low loaded)	V	S	0.0	0.8	0.0	0.8	0.0	0.8	0.0	0.8	V
E1-E19	Input clamp diode voltage	VI	S	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	V
F1-F20	Input current, low level	VII	X	0.00	-0.28	0.00	-0.28	0.00	-0.28	0.00	-0.28	mA
G1-G20	Input current, high level	VII	X	-20	40	-20	40	-20	40	-20	40	µA
H	Power supply current	VIII	X	10	260	10	260	10	260	10	260	mA
J1-J8	Output leakage current (high-impedance state), V_O = 2.4 V	IX	X	-100	100	-100	100	-100	100	-100	100	µA
J9-J16	Output leakage current (high-impedance state), V_O = 0.5 V	IX	X	-100	100	-100	100	-100	100	-100	100	µA
K	Output voltage (V_O) level (unprogrammed)	X	X	← See table X, no failure →								
L	Power supply filter capacitance	XI	X	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
M	Programmability	XII	Q	← See table XII, no failure →								
N	Functional test pattern	XIV	Q	← See table XIV, no failure →								
P1-P8	Output voltage, low level	XV	Q	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
Q1-Q8	Output voltage, high level	XV	Q	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V

See footnotes at end of table

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R1-R8	Transition time (t_{TLH}), C_L = 50 pF	XVI	Q	5	40	5	40	5	40	5	40	ns
R9-R16	Transition time (t_{TLH}), C_L = 150 pF	XVI	Q	5	50	5	50	5	50	5	50	ns
S1-S8	Transition time (t_{THL}), C_L = 50 pF	XVI	Q	3	30	3	30	3	30	3	30	ns
S9-S16	Transition time (t_{THL}), C_L = 150 pF	XVI	Q	5	40	5	40	5	40	5	40	ns
T1-T8	Enable recovery time (t_{ERL}), C_L < 10 pF	XVII	Q	5	40	5	40	5	40	5	40	ns
U1-U8	Enable recovery time (t_{ERH}), C_L < 10 pF	XVII	Q	5	40	5	40	5	40	5	40	ns
V1	Galloping read access pattern (C_L = 50 pF, V_{CC} = 4.5 V)	XVIII	Q	← See table XVIII, no failure →								
V2	Galloping read access pattern (C_L = 50 pF, V_{CC} = 5.5 V)	XVIII	Q	← See table XVIII, no failure →								
V3	Galloping read access pattern (C_L = 150 pF, V_{CC} = 4.5 V)	XVIII	Q	← See table XVIII, no failure →								
V4	Galloping read access pattern (C_L = 150 pF, V_{CC} = 5.5 V)	XVIII	Q	← See table XVIII, no failure →								
W1,W2	Input crosstalk, low level	XIX	Q	0	400	0	400	0	400	0	400	mV
X1,X2	Input crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
Y	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	mV
Z1-Z18	Input capacitance, address	XXI	Q	2	25	2	25	2	25	2	25	pF

See footnotes at end of table

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Z19,Z20	Input capacitance, enable	XXI	Q	2	40	2	40	2	40	2	40	pF
Z21-Z28	Output capacitance	XXI	Q	2	25	2	25	2	25	2	25	pF

1/ See order of testing (4.4.2).
2/ "X" parameters shall be 100 percent tested during module acceptance tests.
3/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
4/ "Q" parameters shall be tested during qualification testing only.

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3.10 Acceptance test. All production modules shall be tested in accordance with footnotes 2 and 3 of table I and as specified in the user's detailed module document.

3.11 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.12 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ± 0.5 V dc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787 and in the user's detailed module document.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. The order of testing for unprogrammed and programmed modules shall be as follows:

4.4.2.1 Unprogrammed module order of testing. Test conditions A through L shall be performed on all unprogrammed modules. Test condition order shall be A, B, C, and D followed by E through L in any order.

4.4.2.2 Programmed module order of testing. Test conditions A through J and L through Z shall be performed on all programmed modules. Test condition order shall be M, A, B, C, and D followed by E through J, L, and N through Z in any order.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modification or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through L in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ± 5°C ambient on all unprogrammed modules. Test conditions A through J, L, and N through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ± 5°C ambient on all programmed modules. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through L and N through Y shall be performed in the order specified in 4.4.2. using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test conditions, N, P, and Q, shall be performed using the initial limits at 25°C ± 5°C ambient.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests E through Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through Y shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

4.6 Freeze-out test. Within no more than 24 hours after completion of burn-in, or if more than 24 hours have elapsed subsequent to the 125°C burn-in exposure, devices shall be conditioned with at least 125° for a minimum of 5 hours immediately prior to the freeze-out test. The freeze-out test shall be conducted as follows:

a. Connect devices in the electrical configuration of test condition C, method 1015 of MIL-STD-883, except that the bias shall be cycled, 3 minutes "on" and 3 minutes "off", throughout the duration of the test.

b. Reduce the device temperature to a maximum of TA of - 10°C and maintain them at that temperature for a minimum of 5 hours duration.

c. With the cycle bias maintained, allow TA to go to room temperature (by removal from the cold chamber or termination of forced cooling but with no forced heating or forced-air flow) and retain for a minimum of 19 hours subsequent to the completion of the 5-hour cold soak. TA shall not exceed 35°C during this period.

d. Remove bias and subject devices to the final electrical tests as specified.

4.7 Nichrome etch test. The nichrome etch test shall be conducted as follows:

a. Add 45 ml deionized water to 10 ml of concentrated nitric acid. Add 6 grams of ceric sulfate $H_4\ Ce(SO_4)_4$ to this mixture and dissolve.

Add sufficient deionized water to make 100 ml of solution. Place a magnetic stirrer in the etchant and heat the solution to 60°C.

b. Immerse the device under test in the solution for 15 seconds while constantly stirring the solution. Rinse in hot (60°C) distilled water and dry.

c. Functionally test the unit at all address locations.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.11).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	1,430 mW (outputs not loaded)
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum; V_{CC} maximum

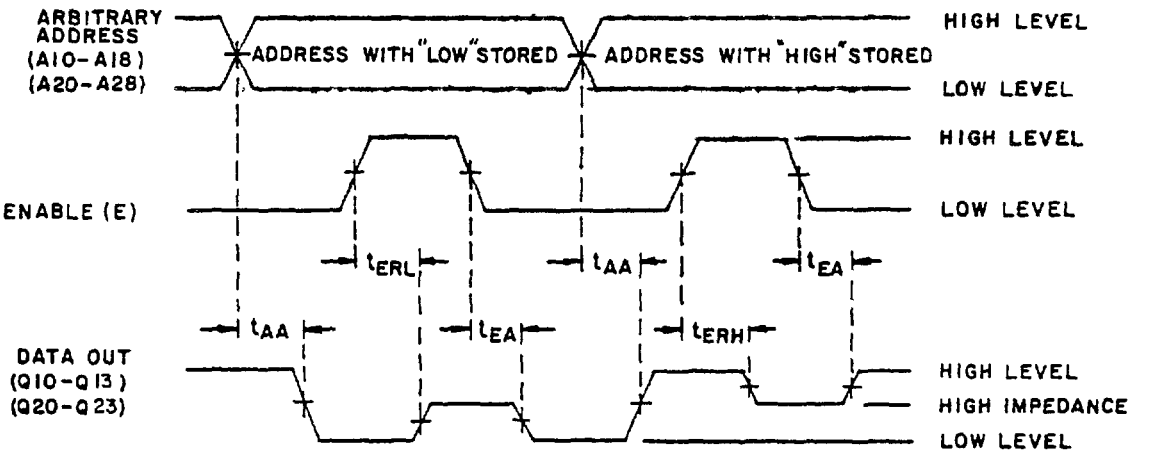
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Input current, address	
Low level (V_{IL} = 0.0 V)	-0.28 mA maximum
(V_{IL} = 0.5 V)	-0.25 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current, enable (program pin)	
Low level (V_{IL} = 0.0 V)	-0.28 mA maximum
(V_{IL} = 0.5 V)	-0.25 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 4.5 V)	1 mA maximum
Input clamp diode voltage	
I_{CLAMP} = -10 mA	-0.2 V minimum, -1.5 V maximum
Output voltage	
Low level (I_{OL} = 16 mA)	0.5 V maximum
High level (I_{OH} = -2 mA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	16 mA
High level (V_{OH} = 2.4 V)	-2 mA
Output short circuit current	
V_{OL} = 0.0 V, V_{CC} = 5.5 V	-15 mA minimum, -90 mA maximum
Output leakage current (high-impedance state)	
$0.5 \leq V_O \leq 2.4$ V	± 100 µA maximum
Propagation delay time (see figure 2)	
t_{AA} - Address to data output	
C_L = 50 pF	80 ns maximum
C_L = 150 pF	90 ns maximum
t_{EA} - Enable	
C_L = 50 pF	45 ns maximum
C_L = 150 pF	55 ns maximum
Enable recovery time	
t_{ERL} - Low-state to high-impedance state (C_L < 10 pF)	40 ns maximum
t_{ERH} - High-state to high-impedance state (C_L < 10 pF)	40 ns maximum
Address input capacitance	25 pF maximum
V_{CC} = 5 V; V_{IL} = 0.5 V; V_{IN} = 15 mV, 1 MHz	
Enable input capacitance	40 pF maximum
V_{CC} = 5 V; V_{IL} = 0.5 V; V_{IN} = 15 mV, 1 MHz	
Output capacitance	25 pF maximum
V_{CC} = 5 V: V_{OL} = 0.5 V; V_{IN} = 15 mV, 1 MHz	

6.4 Module function. The module is organized as two independent 512-word by 4-bit programmable read only memories. These memories may be interconnected through connector wiring to form a memory of either 1024 by 4 or 512 by 8.

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6.4.1 Input to output relationship. The module contains two independent 512-word by 4-bit programmable read only memories (PROM) as shown on figure 1. When the enable of one of the 512-word by 4-bit memories goes to the low level, the data programmed at the selected address input appears at the four data outputs of that memory. When the enable of one of the memories is high, the four output lines of that memory are disabled to a high-impedance state. Module timing and truth table are defined on figure 2.



DYNAMIC CHARACTERISTICS WAVEFORMS

Module truth table.

Enable (E)	Outputs Q10-Q13 (Q20-Q23)
H	High Z
L	Read data of A10-A18 (A20-A28)

H = high-level input
L = low-level input
High Z = high-impedance state

FIGURE 2. Dynamic characteristics waveforms and module truth table.

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6.4.2 Parallel inputs and outputs. Corresponding address inputs of the two address groups must be paralleled for nonindependent operation of the two memory sections. A number of memory outputs can be connected to a common bus. All of the memories except one are placed in the high-impedance state, and the selected memory is enabled. When outputs of several modules are bussed together, all address lines should be addressed using synchronous clocking.

6.4.3 Programming. Module programming technique (see 3.2.2.3) is defined by the GYF module user.

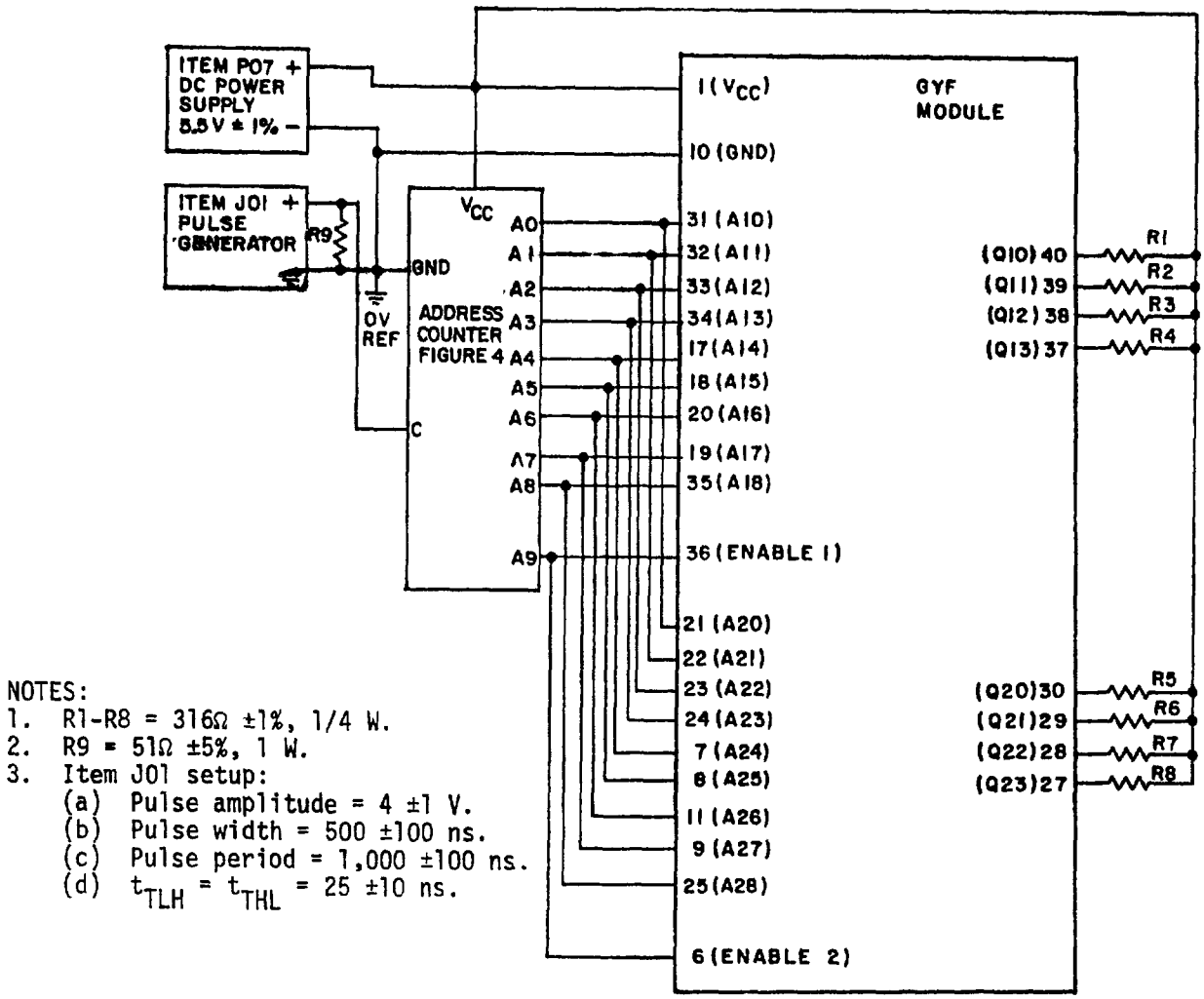
6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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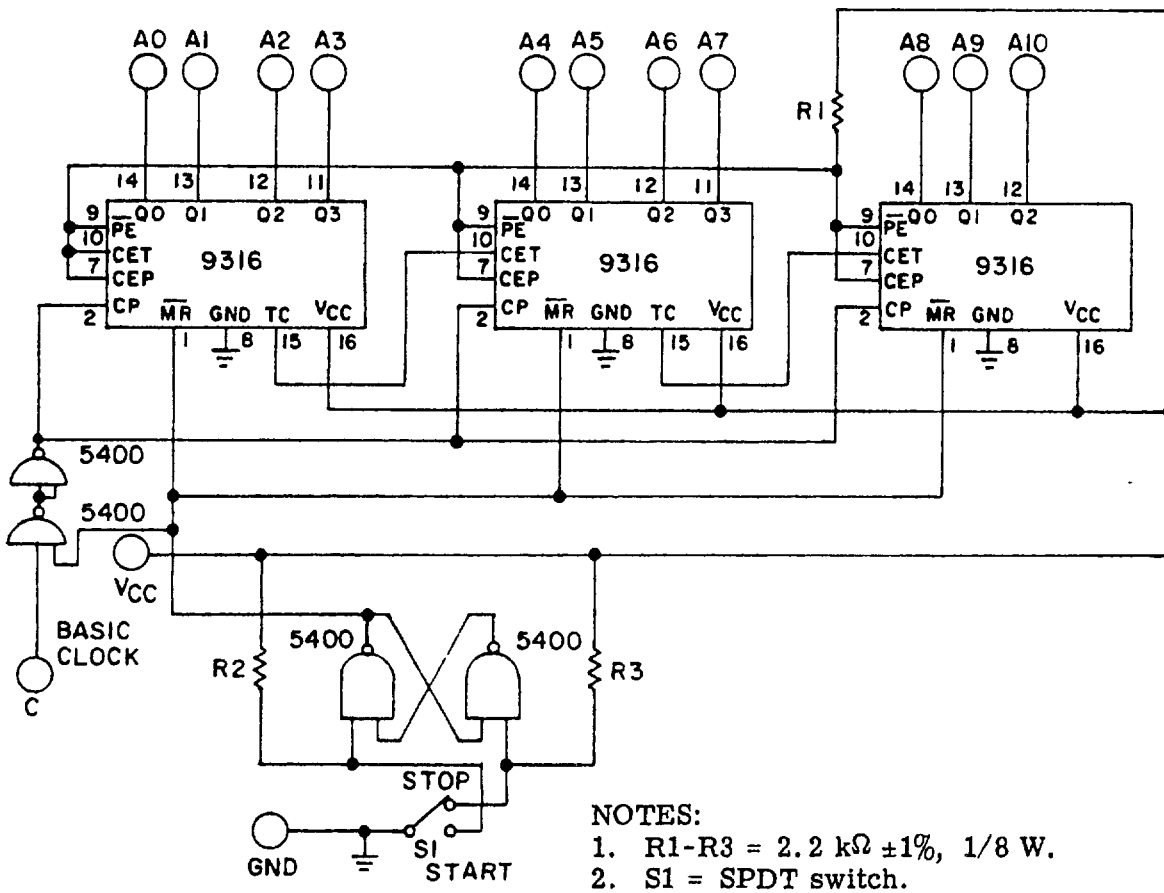
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NOTES:
1. R1-R8 = 316Ω ±1%, 1/4 W.
2. R9 = 51Ω ±5%, 1 W.
3. Item J01 setup:
   (a) Pulse amplitude = 4 ±1 V.
   (b) Pulse width = 500 ±100 ns.
   (c) Pulse period = 1,000 ±100 ns.
   (d) $t_{TLH} = t_{THL}$ = 25 ±10 ns.

FIGURE 3. Operating temperature and life test circuit.

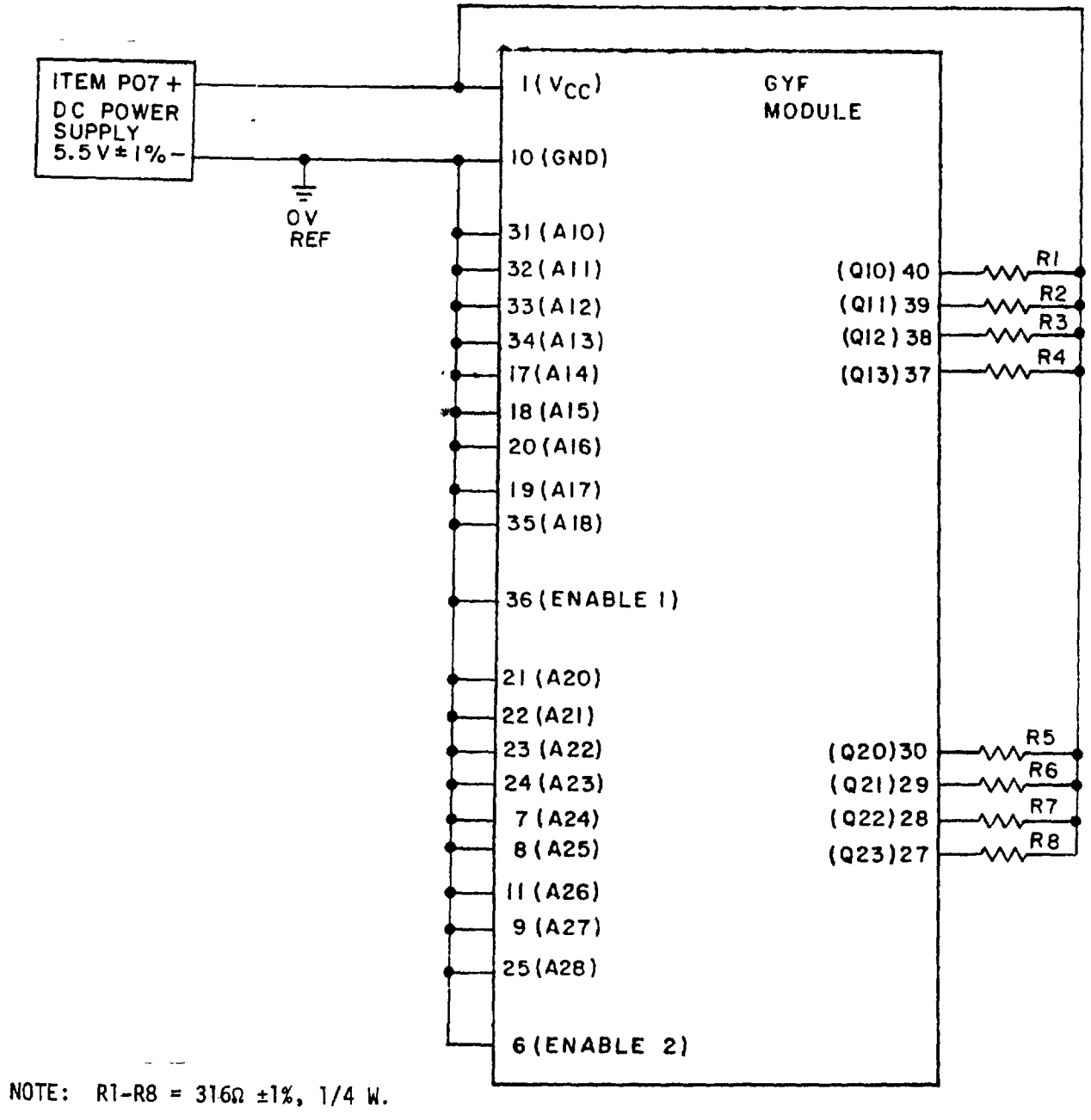
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NOTES:
1. R1-R3 = 2.2 kΩ ±1%, 1/8 W.
2. S1 = SPDT switch.
3. 3 each 9316 high-speed synchronous 4-bit binary counters.
4. 1 each 5400 quad 2-input NAND gate.

FIGURE 4. Address counter circuit.

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NOTE: R1-R8 = 316Ω ±1%, 1/4 W.

FIGURE 5. Component temperature test circuit.

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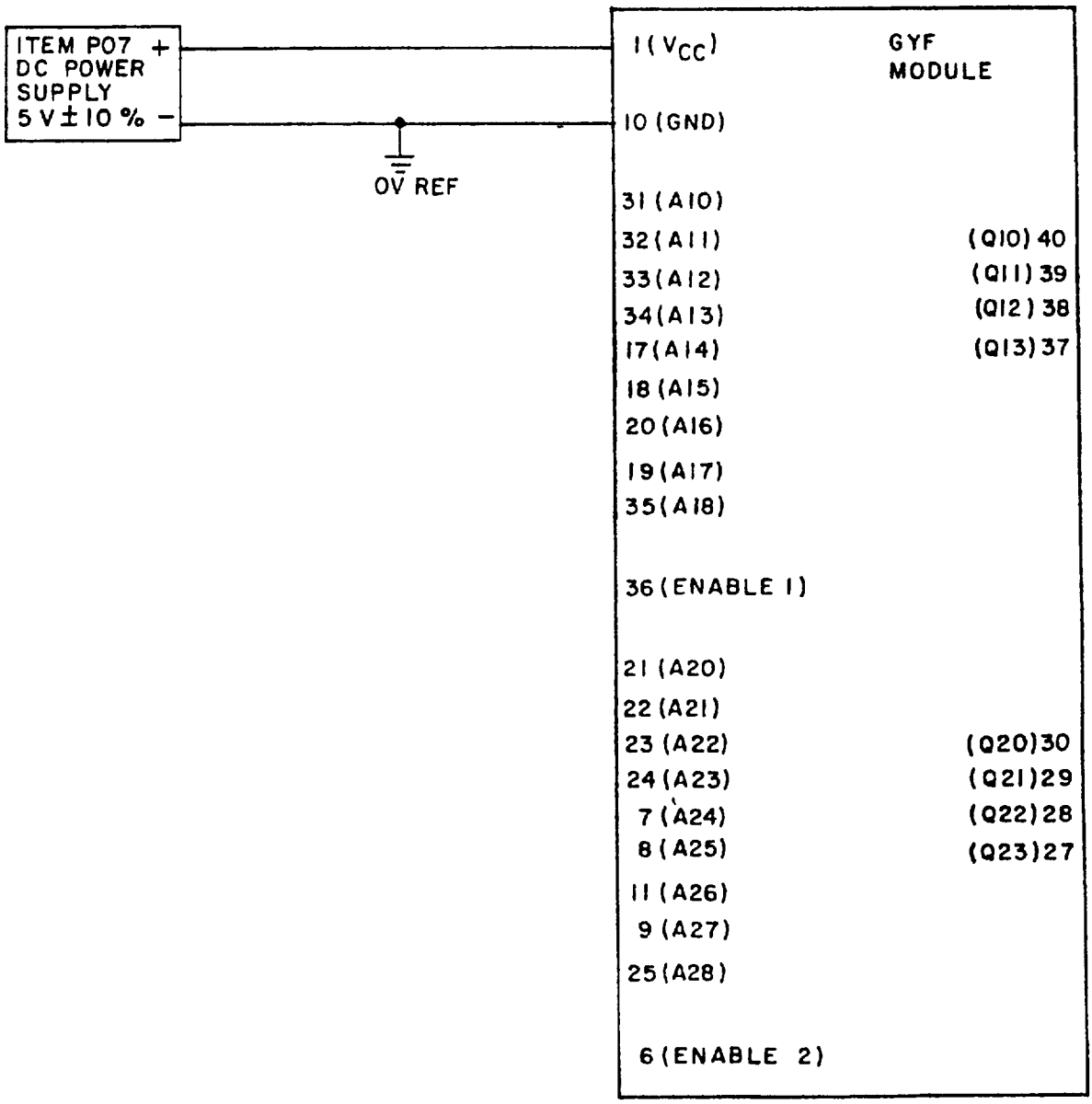


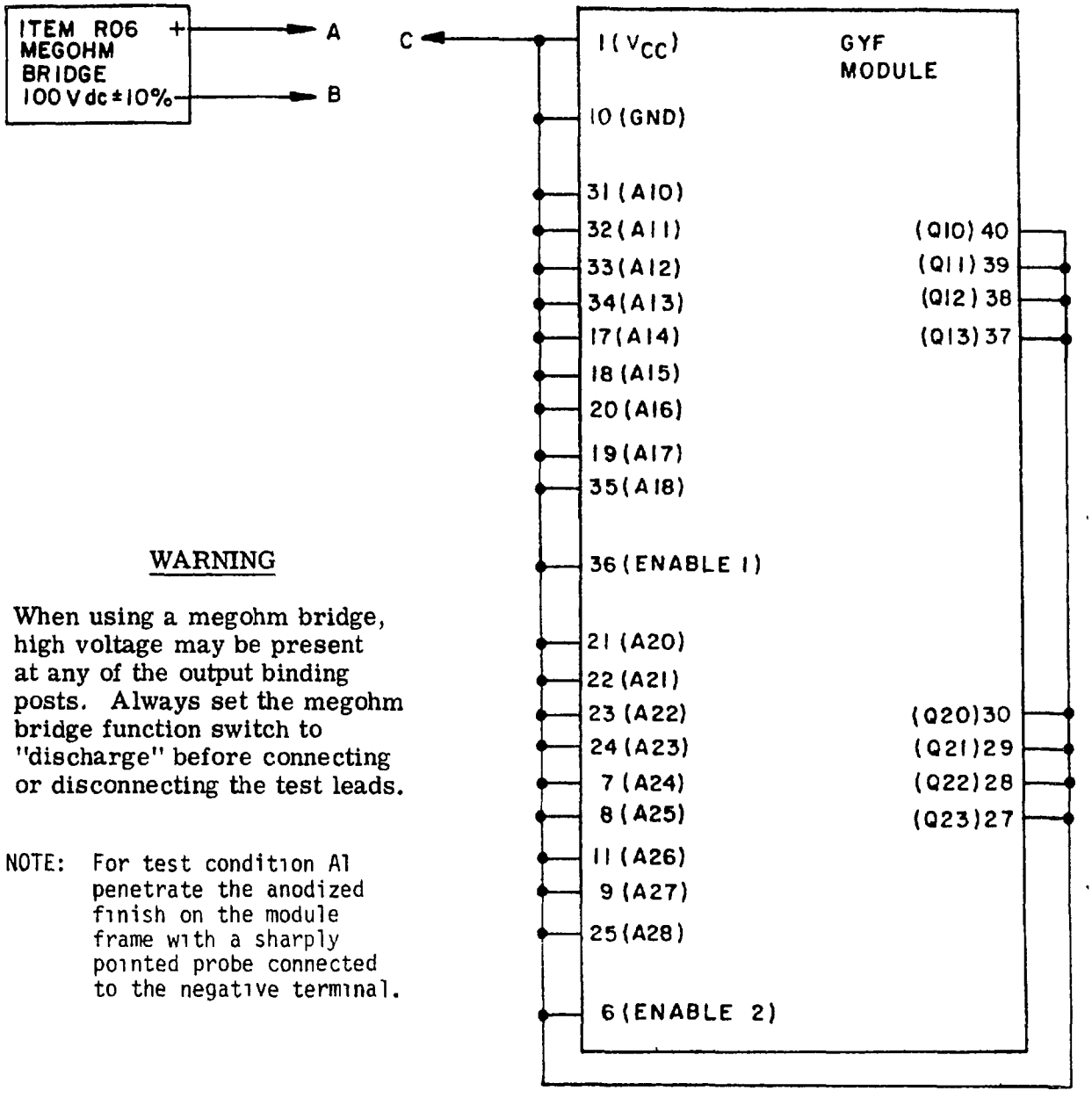
FIGURE 6. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame to module pins	C	Frame	10	∞	MΩ	7
A2	Isolation, unused to used pins	"	2	"	"	"	"
A3	"	"	3	"	"	"	"
A4	"	"	4	"	"	"	"
A5	"	"	5	"	"	"	"
A6	"	"	12	"	"	"	"
A7	"	"	13	"	"	"	"
A8	"	"	14	"	"	"	"
A9	"	"	15	"	"	"	"
A10	"	"	16	"	"	"	"
A11	"	"	26	"	"	"	"

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table II.

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TABLE III. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	A10 Input breakdown current	32,33,34,17,18,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	31	0	1	mA	8
B2	A11 "	31,33,34,17,18,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	32	"	"	"	"
B3	A12 "	31,32,34,17,18,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	33	"	"	"	"
B4	A13 "	31,32,33,17,18,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	34	"	"	"	"
B5	A14 "	31,32,33,34,18,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	17	"	"	"	"
B6	A15 "	31,32,33,34,17,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	18	"	"	"	"
B7	A16 "	31,32,33,34,17,18,19,35,21, 22,23,24,7,8,11,9,25,36,6	20	"	"	"	"
B8	A17 "	31,32,33,34,17,18,20,35,21, 22,23,24,7,8,11,9,25,36,6	19	"	"	"	"
B9	A18 "	31,32,33,34,17,18,20,19,21, 22,23,24,7,8,11,9,25,36,6	35	"	"	"	"
B10	A20 "	31,32,33,34,17,18,20,19,35, 22,23,24,7,8,11,9,25,36,6	21	"	"	"	"
B11	A21 "	31,32,33,34,17,18,20,19,35, 21,23,24,7,8,11,9,25,36,6	22	"	"	"	"
B12	A22 "	31,32,33,34,17,18,20,19,35, 21,22,24,7,8,11,9,25,36,6	23	"	"	"	"
B13	A23 "	31,32,33,34,17,18,20,19,35, 21,22,23,7,8,11,9,25,36,6	24	"	"	"	"
B14	A24 "	31,32,33,34,17,18,20,19,35, 21,22,23,24,8,11,9,25,36,6	7	"	"	"	"

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TABLE III. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B15	A25 Input breakdown current	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,11,9,25,36,6	8	0	1	mA	8
B16	A26 "	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,8,9,25,36,6	11	"	"	"	"
B17	A27 "	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,8,11,25,36,6	9	"	"	"	"
B18	A28 "	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,8,11,9,36,6	25	"	"	"	"
B19	Enable 1 "	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,8,11,9,25,6	36	-0.1	1.0	"	"
B20	Enable 2 "	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,8,11,9,25,36	6	-0.1	1.0	"	"

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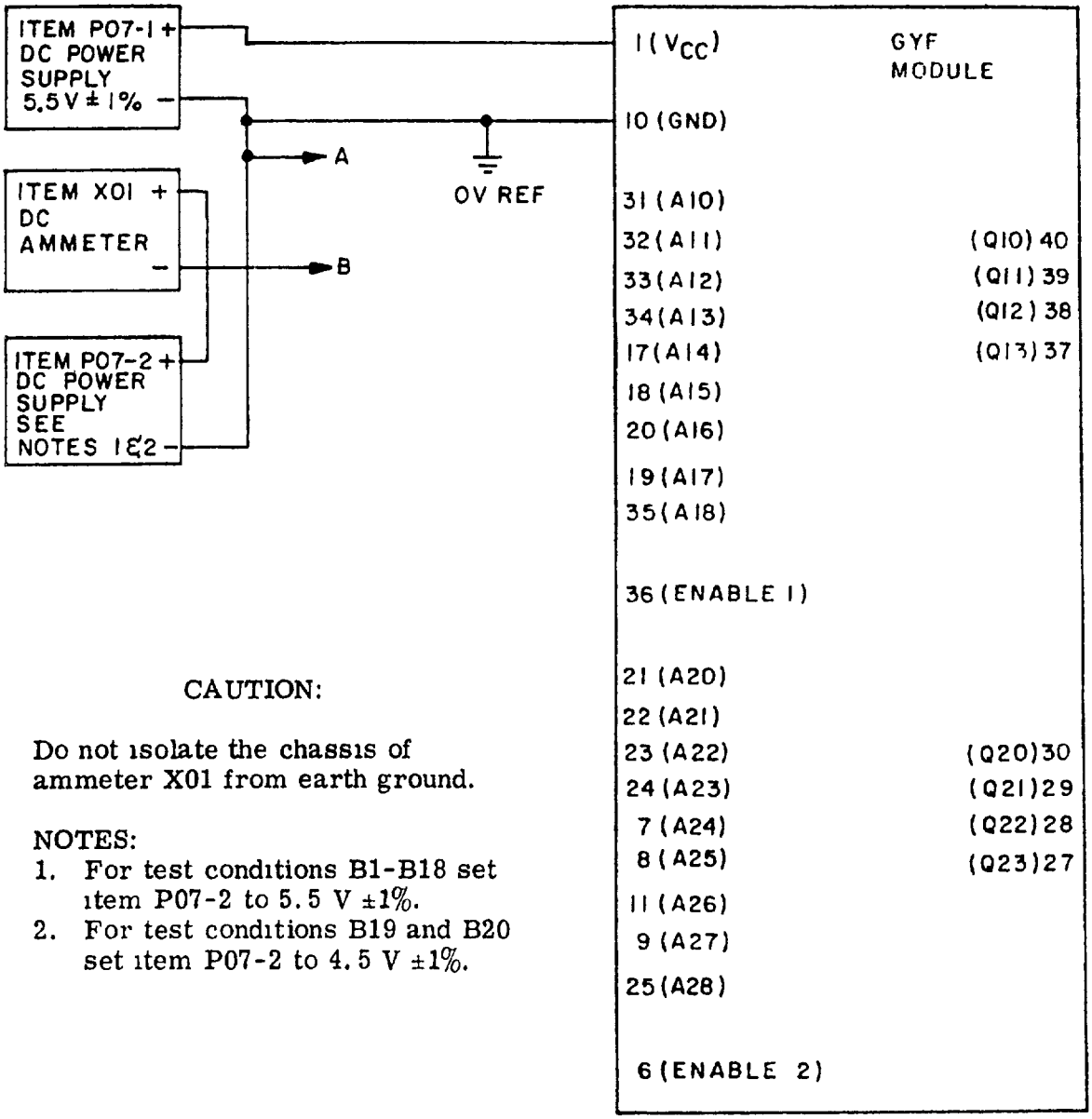
TABLE IV. Output short circuit current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
C1	Q10 Output short circuit current	40	-15	-90	mA	9
C2	Q11 "	39	"	"	"	"
C3	Q12 "	38	"	"	"	"
C4	Q13 "	37	"	"	"	"
C5	Q20 "	30	"	"	"	"
C6	Q21 "	29	"	"	"	"
C7	Q22 "	28	"	"	"	"
C8	Q23 "	27	"	"	"	"

TABLE V. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	Q10 V_{OUT} low loaded	40	0.0	0.8	V	10
D2	Q11 "	39	"	"	"	"
D3	Q12 "	38	"	"	"	"
D4	Q13 "	37	"	"	"	"
D5	Q20 "	30	"	"	"	"
D6	Q21 "	29	"	"	"	"
D7	Q22 "	28	"	"	"	"
D8	Q23 "	27	"	"	"	"

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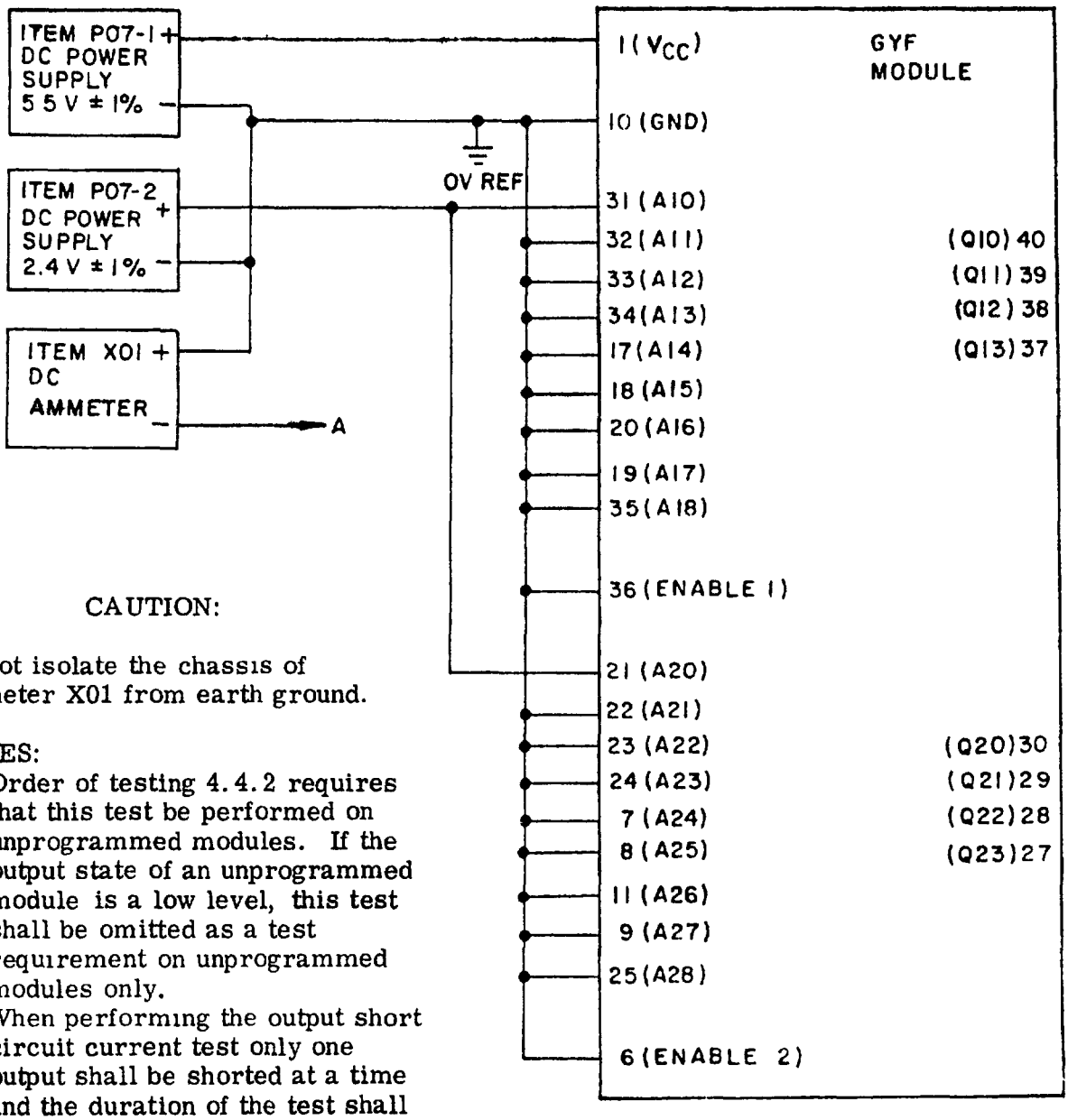
CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. For test conditions B1-B18 set item P07-2 to 5.5 V ±1%.
2. For test conditions B19 and B20 set item P07-2 to 4.5 V ±1%.

FIGURE 8. Input breakdown current test circuit for table III.

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CAUTION:

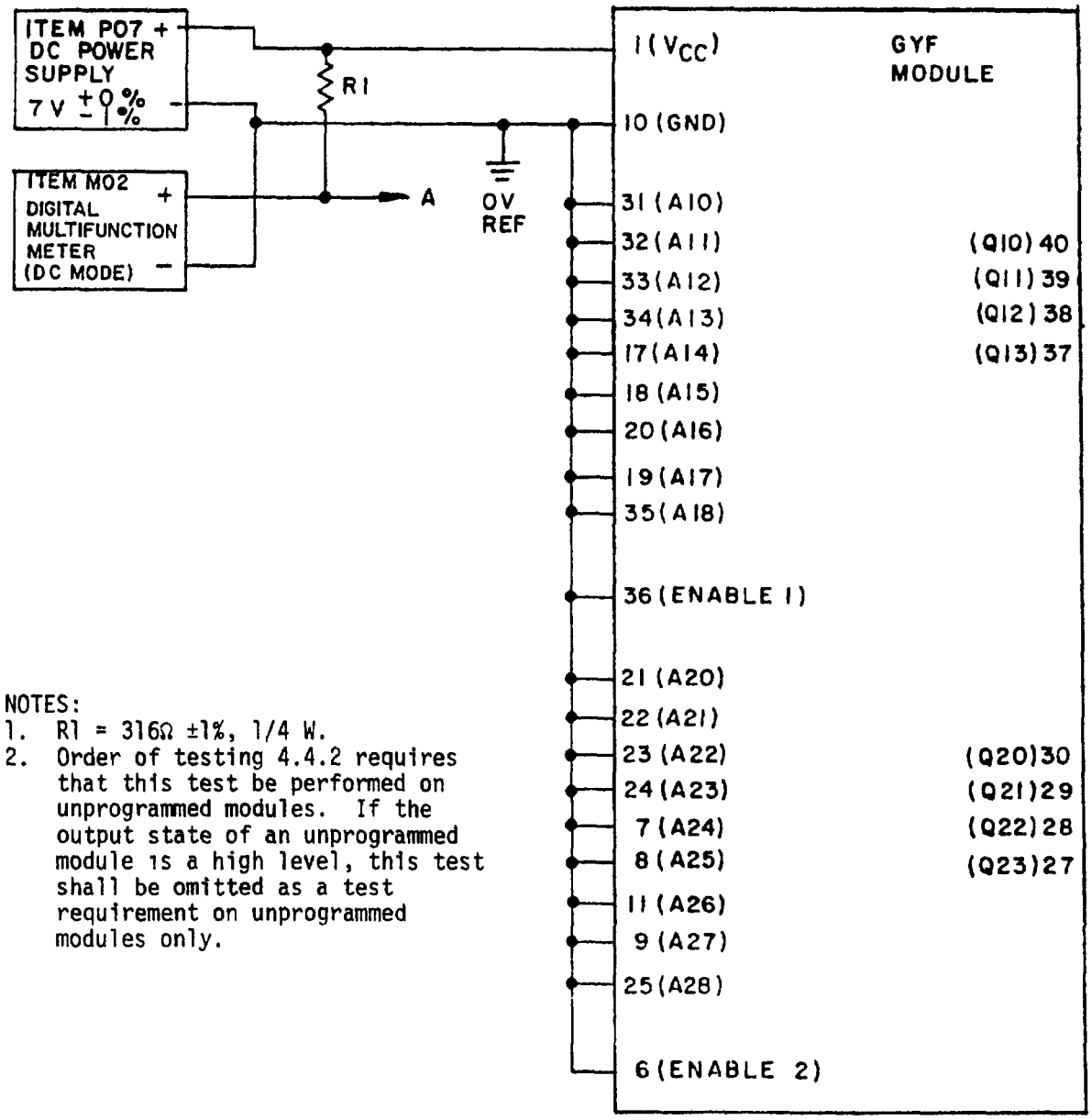
Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:

1. Order of testing 4.4.2 requires that this test be performed on unprogrammed modules. If the output state of an unprogrammed module is a low level, this test shall be omitted as a test requirement on unprogrammed modules only.
2. When performing the output short circuit current test only one output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.

FIGURE 9. Output short circuit current test circuit for table IV.

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NOTES:
1. R1 = 316Ω ±1%, 1/4 W.
2. Order of testing 4.4.2 requires that this test be performed on unprogrammed modules. If the output state of an unprogrammed module is a high level, this test shall be omitted as a test requirement on unprogrammed modules only.

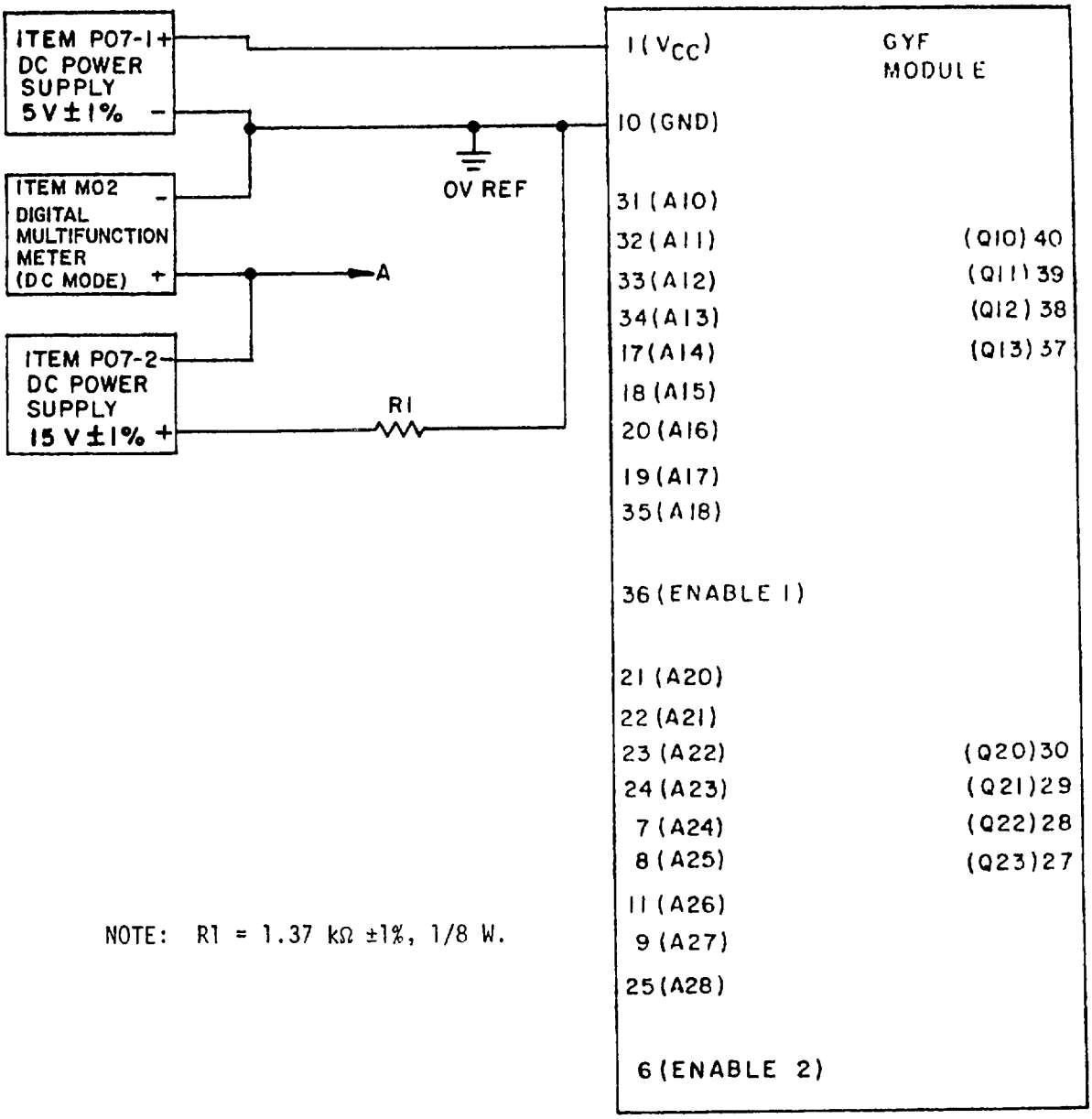
* FIGURE 10. V_{OUT} (low-loaded) test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	A10 Input clamp diode voltage	31	-0.2	-1.5	V	11
E2	A11 "	32	"	"	"	"
E3	A12 "	33	"	"	"	"
E4	A13 "	34	"	"	"	"
E5	A14 "	17	"	"	"	"
E6	A15 "	18	"	"	"	"
E7	A16 "	20	"	"	"	"
E8	A17 "	19	"	"	"	"
E9	A18 "	35	"	"	"	"
E10	A20 "	21	"	"	"	"
E11	A21 "	22	"	"	"	"
E12	A22 "	23	"	"	"	"
E13	A23 "	24	"	"	"	"
E14	A24 "	7	"	"	"	"
E15	A25 "	8	"	"	"	"
E16	A26 "	11	"	"	"	"
E17	A27 "	9	"	"	"	"
E18	A28 "	25	"	"	"	"
E19	Enable 1 "	36	"	"	"	"

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NOTE: R1 = 1.37 kΩ ±1%, 1/8 W.

* FIGURE 11. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Input current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	A10 Input current, low level	Open	31	Open	0.00	-0.28	mA	12
F2	A11 "	"	32	"	"	"	"	"
F3	A12 "	"	33	"	"	"	"	"
F4	A13 "	"	34	"	"	"	"	"
F5	A14 "	"	17	"	"	"	"	"
F6	A15 "	"	18	"	"	"	"	"
F7	A16 "	"	20	"	"	"	"	"
F8	A17 "	"	19	"	"	"	"	"
F9	A18 "	"	35	"	"	"	"	"
F10	A20 "	"	21	"	"	"	"	"
F11	A21 "	"	22	"	"	"	"	"
F12	A22 "	"	23	"	"	"	"	"
F13	A23 "	"	24	"	"	"	"	"
F14	A24 "	"	7	"	"	"	"	"
F15	A25 "	"	8	"	"	"	"	"
F16	A26 "	"	11	"	"	"	"	"
F17	A27 "	"	9	"	"	"	"	"
F18	A28 "	"	25	"	"	"	"	"
F19	Enable 1 "	"	36	"	"	"	"	"
F20	Enable 2 "	"	6	"	"	"	"	"

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TABLE VII. Input current (low-level and high-level) test - Continued.

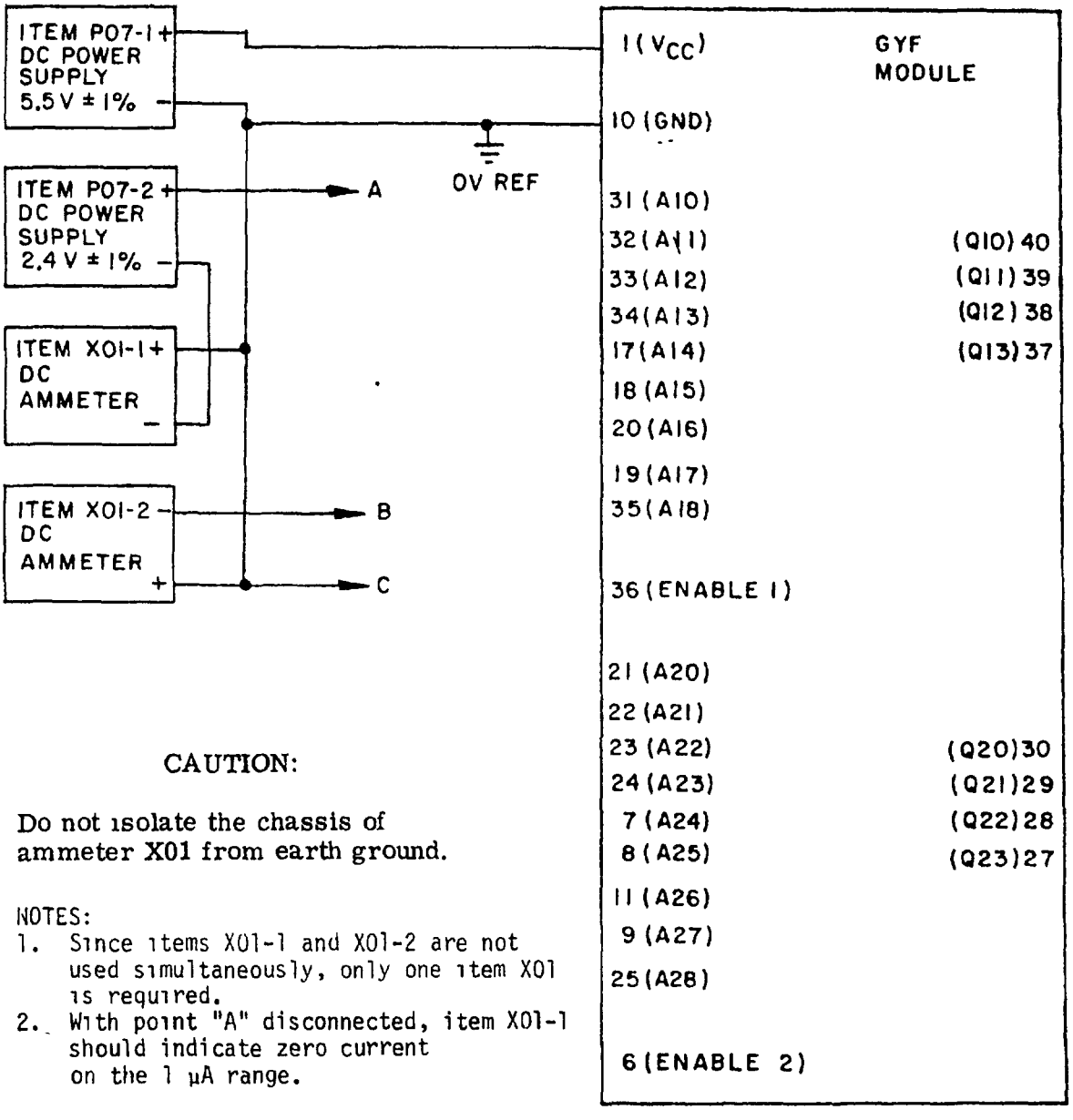
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	A10 Input current, high level	31	Open	32,33,34,17,18,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	-20	40	µA	12
G2	A11 "	32	"	31,33,34,17,18,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	"	"	"	"
G3	A12 "	33	"	31,32,34,17,18,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	"	"	"	"
G4	A13 "	34	"	31,32,33,17,18,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	"	"	"	"
G5	A14 "	17	"	31,32,33,34,18,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	"	"	"	"
G6	A15 "	18	"	31,32,33,34,17,20,19,35,21, 22,23,24,7,8,11,9,25,36,6	"	"	"	"
G7	A16 "	20	"	31,32,33,34,17,18,19,35,21, 22,23,24,7,8,11,9,25,36,6	"	"	"	"
G8	A17 "	19	"	31,32,33,34,17,18,20,35,21, 22,23,24,7,8,11,9,25,36,6	"	"	"	"
G9	A18 "	35	"	31,32,33,34,17,18,20,19,21, 22,23,24,7,8,11,9,25,36,6	"	"	"	"
G10	A20 "	21	"	31,32,33,34,17,18,20,19,35, 22,23,24,7,8,11,9,25,36,6	"	"	"	"
G11	A21 "	22	"	31,32,33,34,17,18,20,19,35, 21,23,24,7,8,11,9,25,36,6	"	"	"	"
G12	A22 "	23	"	31,32,33,34,17,18,20,19,35, 21,22,24,7,8,11,9,25,36,6	"	"	"	"
G13	A23 "	24	"	31,32,33,34,17,18,20,19,35, 21,22,23,7,8,11,9,25,36,6	"	"	"	"
G14	A24 "	7	"	31,32,33,34,17,18,20,19,35, 21,22,23,24,8,11,9,25,36,6	"	"	"	"

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TABLE VII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G15	A25 Input current, high level	8	Open	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,11,9,25,36,6	-20	40	µA	12
G16	A26 "	11	"	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,8,9,25,36,6	"	"	"	"
G17	A27 "	9	"	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,8,11,25,36,6	"	"	"	"
G18	A28 "	25	"	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,8,11,9,36,6	"	"	"	"
G19	Enable 1 "	36	"	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,8,11,9,25,6	"	"	"	"
G20	Enable 2 "	6	"	31,32,33,34,17,18,20,19,35, 21,22,23,24,7,8,11,9,25,36	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. With point "A" disconnected, item X01-1 should indicate zero current on the 1 µA range.

FIGURE 12. Input current (low-level and high-level) test circuit for table VII.

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TABLE VIII. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
H	Power supply current	10	260	mA	13

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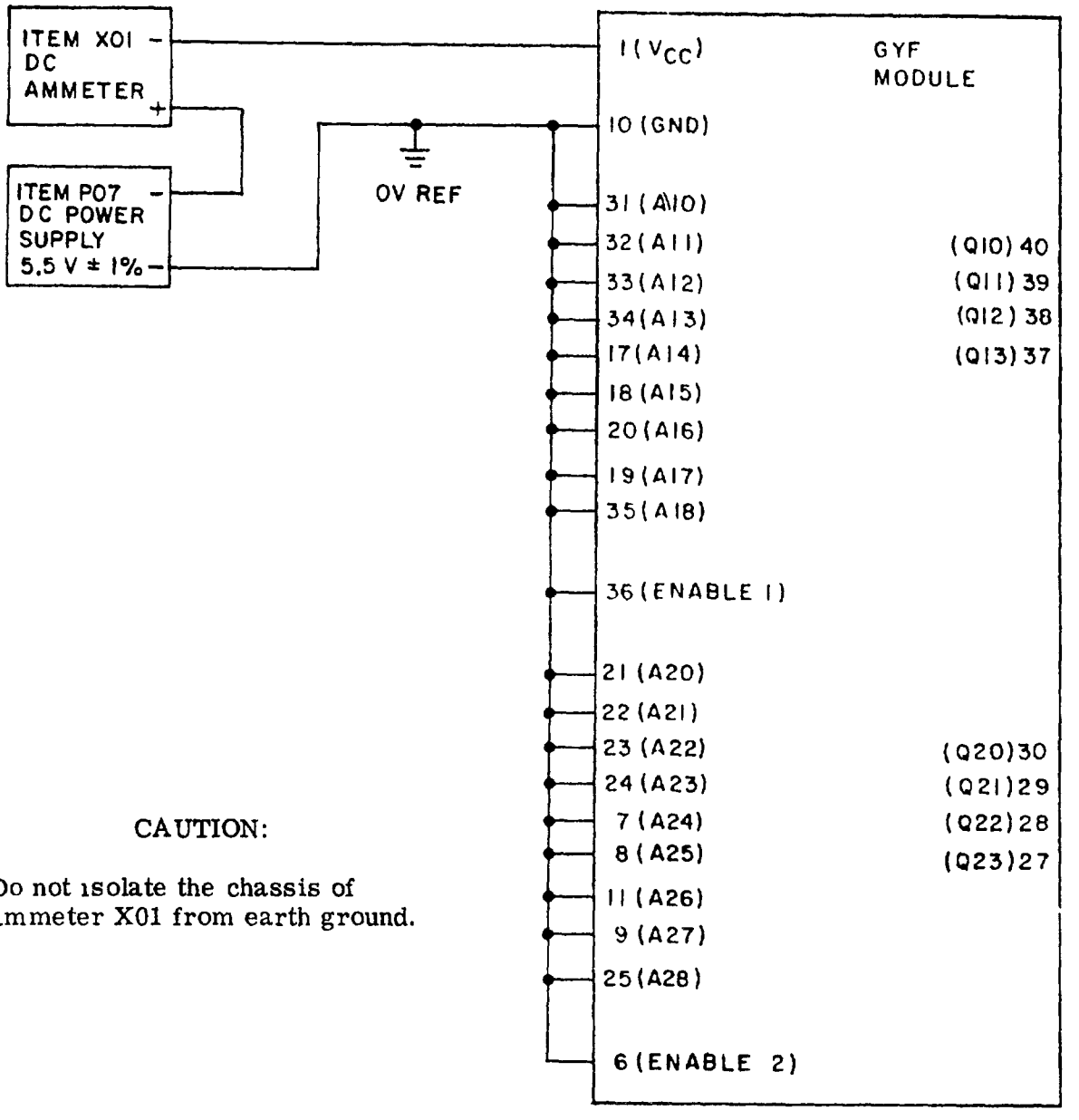


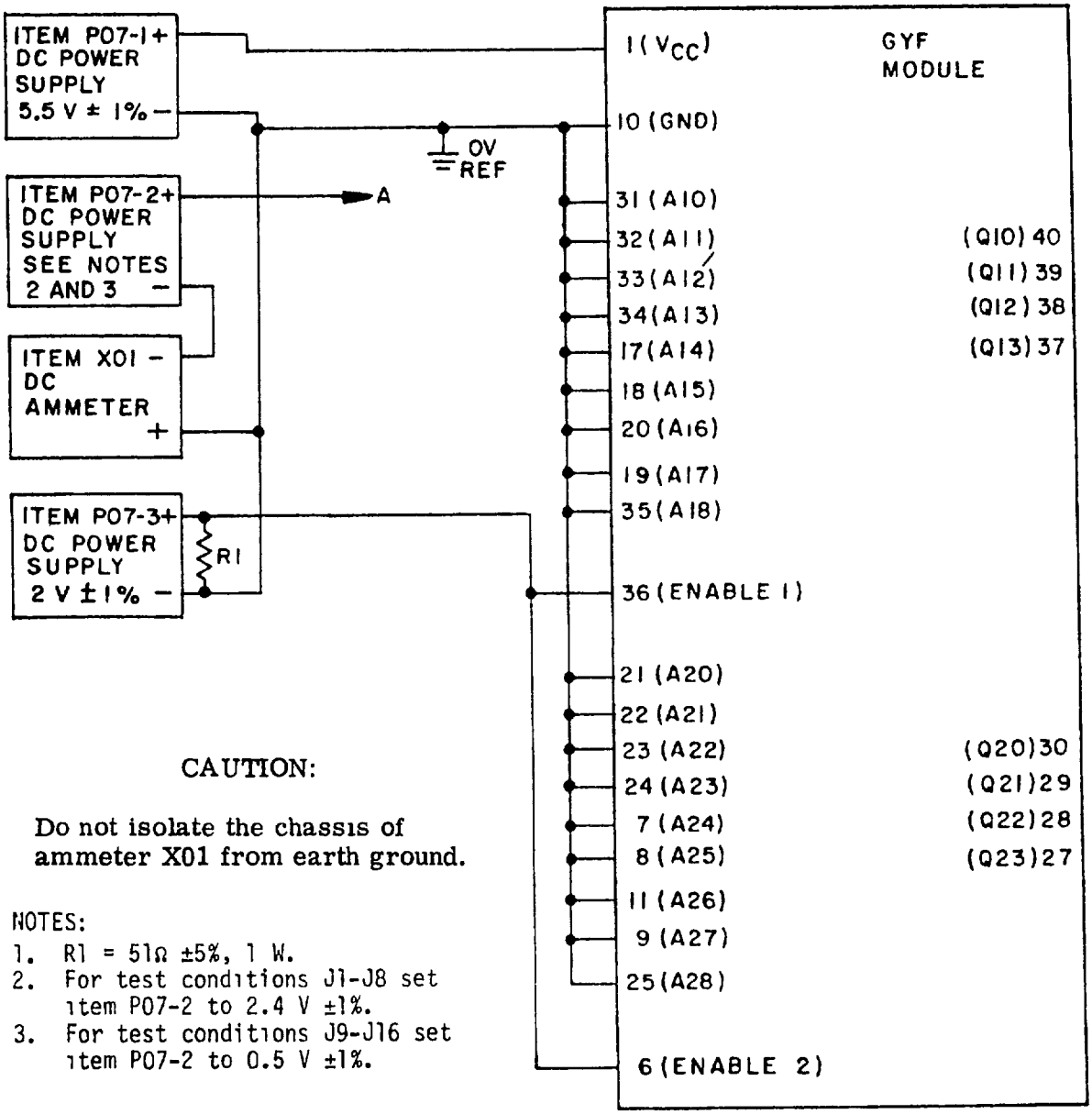
FIGURE 13. Power supply current test circuit for table VIII.

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TABLE IX. Output leakage current (high impedance state) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
J1	Q10	Output leakage current (high-impedance state), V_O = 2.4 V	40	-100	100	µA	14
J2	Q11	"	39	"	"	"	"
J3	Q12	"	38	"	"	"	"
J4	Q13	"	37	"	"	"	"
J5	Q20	"	30	"	"	"	"
J6	Q21	"	29	"	"	"	"
J7	Q22	"	28	"	"	"	"
J8	Q23	"	27	"	"	"	"
J9	Q10	Output leakage current (high-impedance state), V_O = 0.5 V	40	"	"	"	"
J10	Q11	"	39	"	"	"	"
J11	Q12	"	38	"	"	"	"
J12	Q13	"	37	"	"	"	"
J13	Q20	"	30	"	"	"	"
J14	Q21	"	29	"	"	"	"
J15	Q22	"	28	"	"	"	"
J16	Q23	"	27	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. For test conditions J1-J8 set item P07-2 to 2.4 V ±1%.
3. For test conditions J9-J16 set item P07-2 to 0.5 V ±1%.

FIGURE 14. Output leakage current (high-impedance state) test circuit for table IX.

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TABLE X. Output voltage (V_O) level (unprogrammed) test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
K	Output voltage (V_O) level (unprogrammed)	←——See note 1, figure 15——→			

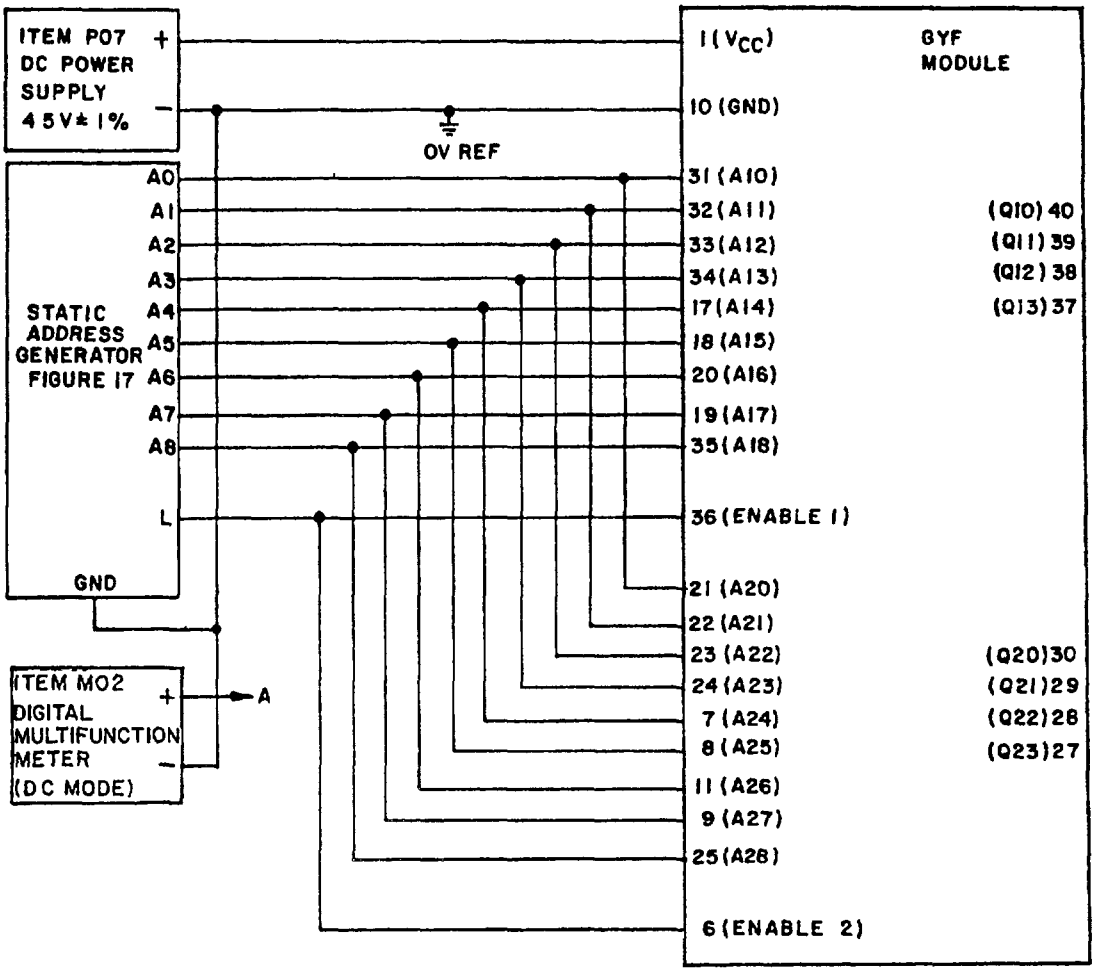
TABLE XI. Power supply filter capacitance test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L	Power supply filter capacitance	1	1.6	3.0	µF	18

TABLE XII. Programmability test.

Test condition	Parameter		Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
M	Programmability	The module shall be programmed in accordance with the user's detailed module document and the test pattern shall be as specified in table XIII. The step by step procedure as specified in the user's detailed module document shall be followed, including the program verification tests.	No failure			

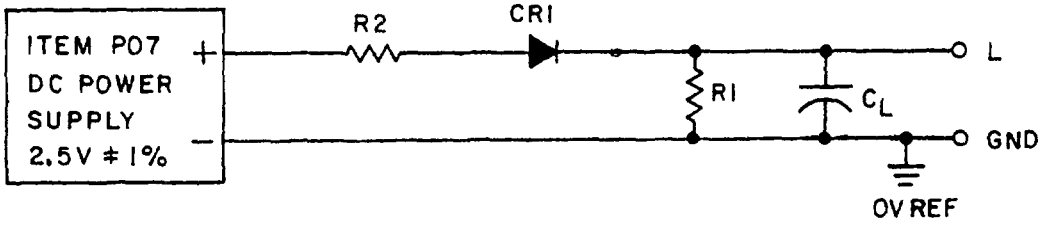
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NOTES:
1. The static address generator is used to select, in turn, each of the 512 address combinations. For each address combination each output is monitored separately at point "A" to verify that all outputs have the voltage level as specified in the user's detailed module document for an unprogrammed module. If an output high level is specified, it shall be $\geq$ 2.4 V, and if a low level is specified, it shall be $\leq$ 0.5 V. An individual module shall have either a high level or a low level for all address combinations.
2. Load the output under test with a load as specified on figure 16.

* FIGURE 15. Output voltage (V_O) level (unprogrammed) test circuit for table X.

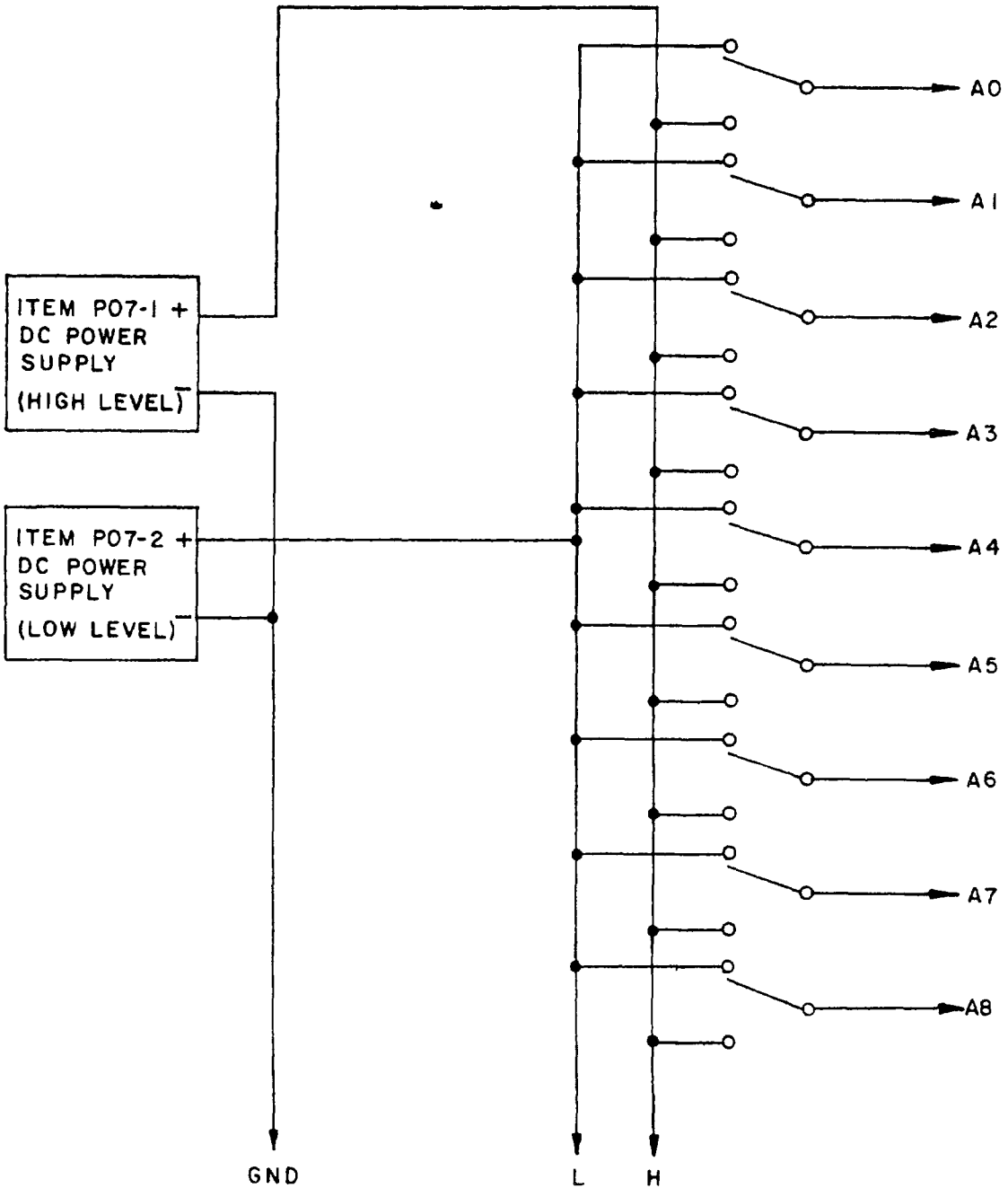
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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = 1.21 kΩ ±1%, 1/8 W.
3. R2 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.5 V.
4. For test conditions R1-R8, S1-S8, V1, V2, W1, W2, X1, and X2, C_L = 50 ±5 pF including probe and test fixture.
5. For test conditions R9-R16, S9-S16, V3, and V4, C_L = 150 ±5 pF including probe and test fixture.
6. C_L is not required for dc electrical tests and test conditions T1-T8 and U1-U8.
7. During output switching the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 16. GYF logic load circuit.

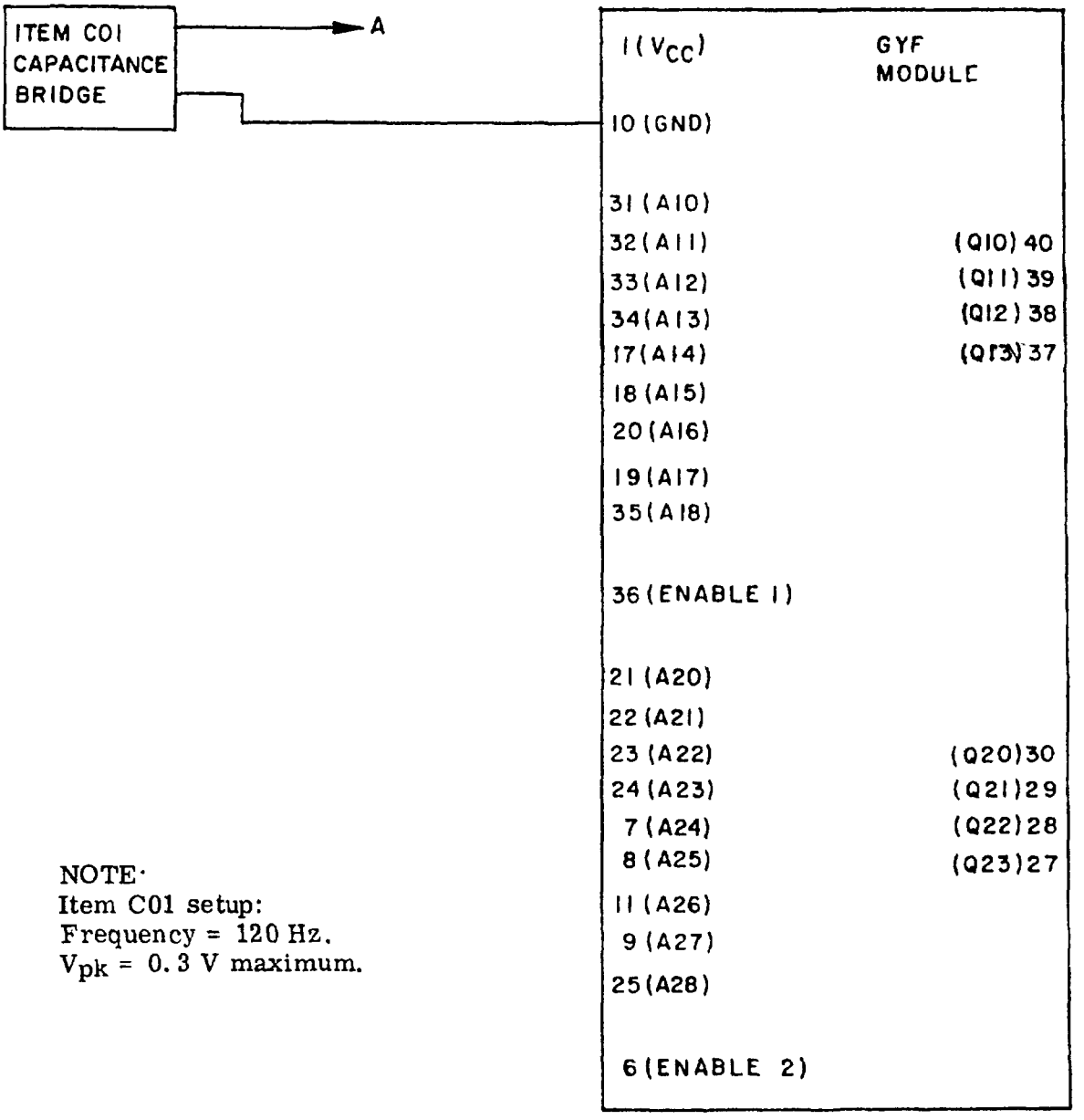
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NOTES:
1. High level (H) is defined as 2.0 V ±1%.
2. Low level (L) is defined as 0.8 V ±1%.

FIGURE 17. Static address generator.

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NOTE:
Item C01 setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

FIGURE 18. Power supply filter capacitance test circuit for table XI.

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TABLE XIII. Memory qualification test pattern. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
000	L	L	L	L	022	H	H	L	L
001	H	H	H	H	023	L	L	H	H
002	L	L	L	H	024	L	H	L	H
003	L	L	H	H	025	H	L	H	L
004	L	L	H	L	026	L	L	H	H
005	L	H	L	H	027	L	H	H	L
006	L	H	H	L	028	L	L	L	L
007	H	L	L	H	029	H	H	L	L
008	L	L	H	H	030	H	L	L	H
009	H	L	H	L	031	L	H	L	H
010	H	H	L	L	032	L	H	L	H
011	L	H	L	H	033	L	L	H	H
012	L	L	L	L	034	L	H	H	L
013	H	L	L	H	035	H	L	H	L
014	H	L	H	L	036	L	L	L	L
015	L	H	L	H	037	L	H	L	H
016	L	H	L	L	038	H	H	L	L
017	H	H	H	H	039	L	L	H	H
018	H	L	H	L	040	L	H	L	H
019	H	H	L	L	041	L	H	H	L
020	L	L	L	L	042	L	L	L	L
021	L	H	H	L	043	H	L	L	H

See footnotes at end of table.

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TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
044	H	L	H	L	066	H	L	L	H
045	H	H	L	L	067	H	L	H	L
046	L	L	H	H	068	H	H	L	L
047	H	H	H	H	069	L	L	H	H
048	H	H	H	H	070	L	H	L	H
049	L	H	L	H	071	L	L	L	L
050	L	L	L	L	072	H	L	L	H
051	L	H	H	L	073	H	H	L	L
052	H	L	L	H	074	L	H	L	H
053	H	L	H	L	075	H	L	H	L
054	H	H	L	L	076	L	L	H	H
055	L	L	H	H	077	H	H	L	L
056	L	H	L	H	078	H	H	H	H
057	L	H	H	L	079	L	L	L	L
058	L	L	L	L	080	L	H	L	H
059	H	L	L	H	081	H	L	H	L
060	H	L	H	L	082	H	H	H	H
061	H	H	L	L	083	H	L	L	H
062	L	L	H	H	084	H	H	L	L
063	L	H	L	H	085	L	L	L	L
064	L	H	H	L	086	L	L	H	H
065	L	H	H	L	087	L	H	H	L

See footnotes at end of table.

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TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
088	H	L	H	L	110	L	L	H	H
089	L	H	L	H	111	L	H	H	L
090	H	L	L	H	112	L	H	L	H
091	H	H	L	L	113	L	L	L	L
092	L	H	H	L	114	H	L	H	L
093	L	L	L	L	115	H	H	H	H
094	L	L	H	H	116	H	H	L	L
095	L	H	L	H	117	L	H	H	L
096	L	H	H	L	118	L	L	H	H
097	H	L	H	L	119	L	H	H	L
098	H	H	L	L	120	H	H	L	L
099	L	L	L	L	121	L	L	L	L
100	L	H	L	H	122	L	H	L	H
101	H	L	L	H	123	H	L	L	H
102	H	L	H	L	124	H	L	H	L
103	L	H	L	H	125	L	L	H	H
104	L	H	H	L	126	H	L	L	H
105	H	L	L	H	127	L	H	L	H
106	H	L	H	L	128	L	H	H	H
107	L	L	L	L	129	H	L	H	L
108	H	H	L	L	130	H	H	L	L
109	H	H	H	H	131	L	H	L	H

See footnotes at end of table.

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TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
132	L	L	H	H	154	H	L	L	H
133	H	L	L	H	155	L	H	L	H
134	L	L	L	L	156	L	H	H	L
135	L	H	H	L	157	H	L	H	L
136	H	H	L	L	158	L	L	L	L
137	L	H	L	H	159	H	H	L	L
138	L	H	H	L	160	L	L	H	H
139	H	L	H	L	161	L	H	L	H
140	H	H	H	H	162	L	H	H	L
141	L	H	H	L	163	H	L	L	H
142	L	L	L	L	164	H	L	H	L
143	H	H	L	L	165	H	H	L	L
144	H	L	L	H	166	L	H	L	H
145	L	L	H	H	167	L	L	L	L
146	H	L	L	H	168	H	L	L	H
147	L	H	H	L	169	L	L	H	H
148	H	H	H	H	170	L	H	H	L
149	H	H	L	L	171	H	H	H	H
150	L	L	L	L	172	H	L	H	L
151	L	H	L	H	173	H	H	L	L
152	H	L	H	L	174	L	H	H	L
153	L	L	H	H	175	L	L	L	L

See footnotes at end of table.

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TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
176	L	H	L	H	198	L	L	L	L
177	H	L	H	L	199	L	L	H	H
178	L	L	H	H	200	H	H	L	L
179	H	L	L	H	201	L	H	L	H
180	H	H	L	L	202	H	H	H	H
181	H	H	H	H	203	H	L	L	H
182	L	L	H	H	204	H	L	H	L
183	L	L	L	L	205	L	H	H	L
184	L	H	H	L	206	L	L	L	L
185	H	L	H	L	207	H	H	L	L
186	L	H	L	H	208	L	H	H	L
187	H	L	L	H	209	L	H	L	H
188	L	H	H	L	210	L	L	H	H
189	H	H	L	L	211	H	H	L	L
190	L	L	H	H	212	L	L	L	L
191	L	L	L	L	213	H	L	H	L
192	L	H	L	H	214	H	H	H	H
193	L	H	H	L	215	H	L	L	H
194	H	L	L	H	216	L	H	L	H
195	H	H	L	L	217	L	H	H	L
196	L	H	H	L	218	H	L	L	H
197	H	L	H	L	219	H	L	H	L

See footnotes at end of table.

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TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
220	L	L	L	L	242	H	L	L	H
221	L	L	H	H	243	L	H	L	H
222	H	H	L	L	244	H	L	H	L
223	L	H	L	H	245	H	H	L	L
224	H	L	L	H	246	H	L	L	H
225	H	H	L	L	247	H	H	H	H
226	L	L	L	L	248	L	L	L	L
227	L	L	H	H	249	L	L	H	H
228	L	H	H	L	250	L	H	H	L
229	H	L	H	L	251	H	L	H	L
230	L	H	L	H	252	L	H	L	H
231	H	L	L	H	253	H	H	L	L
232	L	H	H	L	254	H	L	L	H
233	H	H	H	H	255	L	L	H	H
234	L	L	L	L	256	H	L	L	L
235	L	L	H	H	257	L	H	H	L
236	H	L	H	L	258	H	H	L	L
237	H	H	L	L	259	L	H	L	H
238	L	H	H	L	260	H	L	L	H
239	H	L	H	L	261	H	L	H	L
240	L	L	L	L	262	L	L	H	H
241	L	L	H	H	263	L	L	L	L

See footnotes at end of table.

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TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
264	H	H	H	H	286	L	H	H	L
265	H	H	L	L	287	L	L	H	H
266	L	H	L	H	288	H	L	H	L
267	L	H	H	L	289	L	H	L	H
268	H	L	L	H	290	L	H	H	L
269	H	L	H	L	291	L	L	L	L
270	L	L	H	H	292	L	L	H	H
271	L	L	L	L	293	H	L	H	L
272	H	H	L	L	294	H	H	L	L
273	L	H	H	L	295	H	H	H	H
274	H	L	H	L	296	L	H	H	L
275	L	H	L	H	297	L	H	L	H
276	H	L	L	H	298	H	L	L	H
277	L	L	L	L	299	L	L	L	L
278	L	H	H	L	300	L	L	H	H
279	L	L	H	H	301	H	L	L	H
280	H	H	H	H	302	L	H	L	H
281	H	H	H	H	303	L	H	H	L
282	H	L	H	L	304	H	H	L	L
283	H	H	L	L	305	L	L	L	L
284	L	H	L	H	306	H	L	H	L
285	L	L	L	L	307	L	L	H	H

See footnotes at end of table.

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TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
308	H	L	L	H	330	H	L	L	H
309	L	H	L	H	331	L	L	L	L
310	H	L	H	L	332	L	L	H	H
311	L	H	H	L	333	H	H	L	L
312	H	H	L	L	334	L	H	L	H
313	L	L	L	L	335	H	L	H	L
314	H	H	H	H	336	L	L	L	L
315	L	L	H	H	337	L	H	H	L
316	H	L	H	L	338	H	L	L	H
317	L	H	L	H	339	L	L	H	H
318	L	H	H	L	340	L	H	L	H
319	H	L	L	H	341	L	H	H	L
320	H	H	L	L	342	H	L	L	H
321	H	L	L	H	343	H	L	H	L
322	L	H	H	L	344	L	L	L	L
323	L	L	L	L	345	L	L	H	H
324	L	L	H	H	346	L	H	L	H
325	L	H	H	L	347	H	H	H	H
326	H	H	H	H	348	H	H	H	H
327	H	L	L	H	349	L	L	H	H
328	H	L	H	L	350	H	L	L	H
329	H	H	L	L	351	L	H	H	L

See footnotes at end of table.

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TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
352	L	L	L	L	374	L	H	L	H
353	H	H	L	L	375	H	L	L	H
354	H	L	H	L	376	H	H	L	L
355	L	L	H	H	377	H	L	H	L
356	L	H	H	L	378	L	L	H	H
357	H	H	H	H	379	L	H	L	H
358	L	H	L	H	380	L	L	L	L
359	H	L	H	L	381	H	H	H	H
360	L	L	L	L	382	H	L	L	H
361	L	H	H	L	383	H	L	H	L
362	H	L	L	H	384	L	H	H	L
363	H	H	L	L	385	H	H	L	L
364	H	L	H	L	386	H	L	L	H
365	L	H	L	H	387	L	H	L	H
366	L	H	H	L	388	H	H	H	H
367	L	L	H	H	389	L	H	H	L
368	H	L	L	H	390	L	L	L	L
369	H	H	L	L	391	H	H	L	L
370	L	L	H	H	392	H	L	H	L
371	H	L	H	L	393	L	L	H	H
372	L	L	L	L	394	L	H	L	H
373	L	H	H	L	395	L	H	H	L

See footnotes at end of table.

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TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
396	H	L	H	L	418	L	L	L	L
397	H	L	L	H	419	H	H	H	H
398	L	L	L	L	420	L	H	L	H
399	H	H	L	L	421	H	L	L	H
400	L	L	H	H	422	H	H	L	L
401	H	L	H	L	423	H	L	H	L
402	H	L	L	H	424	L	L	H	H
403	L	H	L	H	425	H	L	L	H
404	L	L	L	L	426	L	L	L	L
405	L	H	L	H	427	L	H	L	H
406	H	L	L	H	428	L	H	H	L
407	H	H	L	L	429	H	H	L	L
408	L	L	H	H	430	H	L	H	L
409	L	H	L	H	431	L	H	H	L
410	L	H	H	L	432	L	L	L	L
411	H	L	L	H	433	L	L	H	H
412	L	L	L	L	434	L	H	L	H
413	H	L	H	L	435	L	H	H	L
414	H	H	H	H	436	H	L	L	H
415	H	H	H	H	437	L	H	H	L
416	L	L	H	H	438	H	H	L	L
417	L	H	H	L	439	L	H	L	H

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:24Z
Check the source to verify that this is the current version before use.

TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
440	L	L	L	L	462	H	L	H	L
441	L	L	H	H	463	H	L	L	H
442	H	L	H	L	464	L	L	H	H
443	L	H	L	H	465	L	L	L	L
444	L	H	H	L	466	L	H	H	L
445	H	L	H	L	467	H	L	L	H
446	H	L	L	H	468	H	H	L	L
447	H	H	L	L	469	H	L	H	L
448	L	L	H	H	470	L	H	L	H
449	L	L	L	L	471	H	L	L	H
450	H	H	H	H	472	L	H	H	L
451	L	H	L	H	473	L	L	L	L
452	H	L	L	H	474	L	L	H	H
453	L	L	H	H	475	H	L	L	H
454	L	H	H	L	476	H	L	H	L
455	H	L	H	L	477	L	H	H	L
456	H	H	L	L	478	L	H	L	H
457	L	L	L	L	479	H	H	L	L
458	L	L	H	H	480	H	H	H	H
459	L	H	H	L	481	L	L	L	L
460	L	H	L	H	482	L	L	H	H
461	H	H	L	L	483	L	H	H	L

See footnotes at end of table.

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TABLE XIII. Memory qualification test pattern - Continued. 1/ - 3/

Address word	Memory outputs				Address word	Memory outputs			
	Q0	Q1	Q2	Q3		Q0	Q1	Q2	Q3
484	L	H	L	H	498	L	H	H	L
485	H	L	H	L	499	H	L	L	H
486	H	H	L	L	500	H	L	H	L
487	H	L	L	H	501	H	H	L	L
488	L	L	L	L	502	L	H	L	H
489	L	L	H	H	503	H	L	L	H
490	L	H	L	H	504	L	L	L	L
491	L	H	H	L	505	L	L	H	H
492	H	L	L	H	506	L	H	H	L
493	H	L	H	L	507	H	H	L	L
494	H	H	L	L	508	H	L	H	L
495	L	L	H	H	509	L	H	L	H
496	L	L	L	L	510	H	L	L	H
497	L	H	L	H	511	L	H	H	L

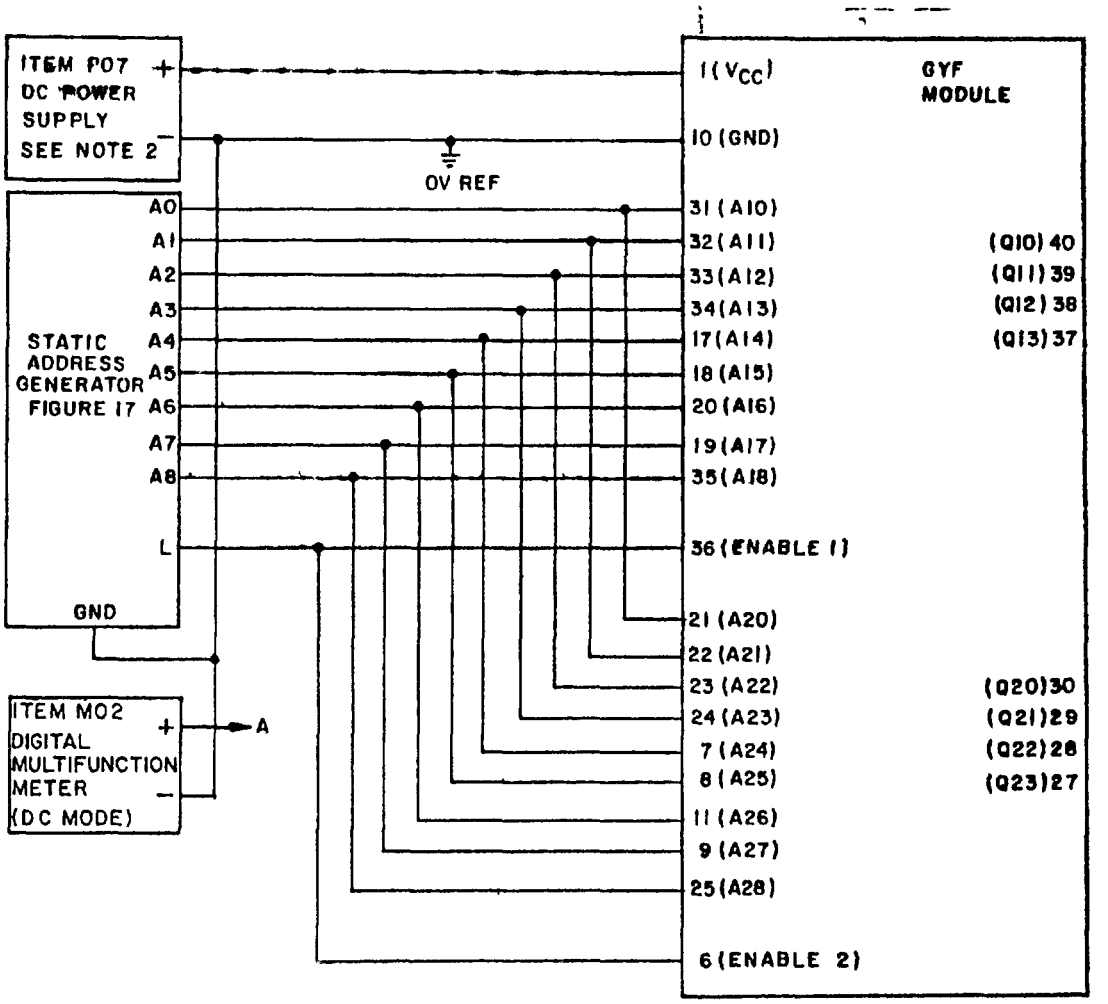
1/ Program each of the two independent 512-word by 4-bit PROM with this test pattern.
2/ L = Low-level output.
3/ H = High-level output.

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TABLE XIV. Functional test pattern test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
N	Functional test pattern	← See note 1, figure 19 →			

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NOTES:
1. The static address generator is used to select, in turn, each of the 512 address combinations. For each address combination the output is monitored at point "A" to verify the test pattern of table XIII. Output high level shall be $\geq$ 2.4 V and output low level shall be $\leq$ 0.5 V.
2. Adjust item P07 either as specified in the user's detailed module document or, if not specified, to 4.5 V ±1%.
3. Load the output under test either with the load as specified in the user's detailed module document or, if not specified, with the load as specified on figure 16.
4. The purpose of notes 2 and 3 is to allow the user to select a V_{CC} voltage and a load current which will insure the PROM fuse link has been blown.

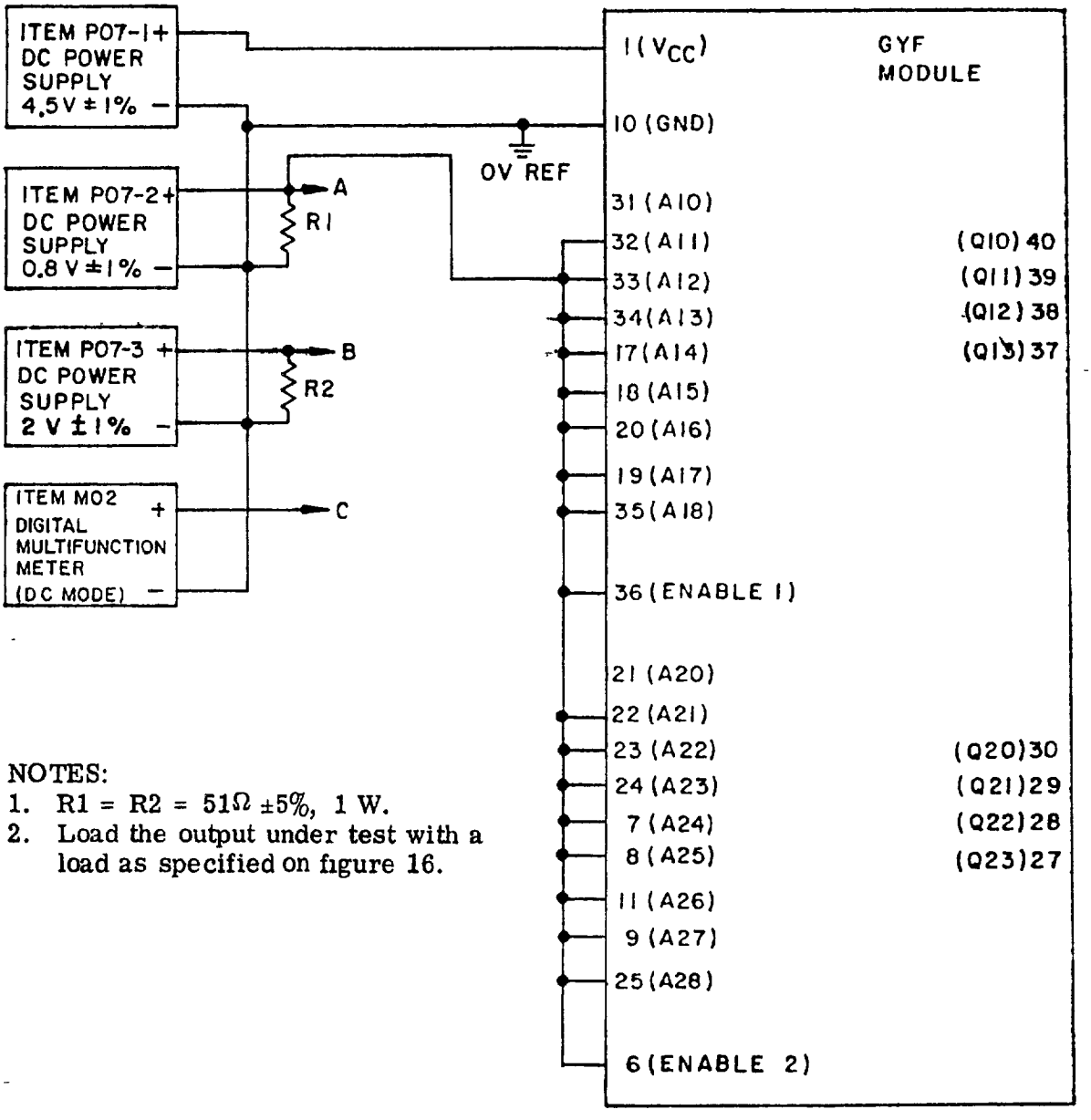
FIGURE 19. Functional test pattern test circuit for table XIV.

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TABLE XV. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Q10 Output voltage, low level	31	Open	40	0.0	0.5	V	20
P2	Q11 "	"	"	39	"	"	"	"
P3	Q12 "	"	"	38	"	"	"	"
P4	Q13 "	"	"	37	"	"	"	"
P5	Q20 "	21	"	30	"	"	"	"
P6	Q21 "	"	"	29	"	"	"	"
P7	Q22 "	"	"	28	"	"	"	"
P8	Q23 "	"	"	27	"	"	"	"
Q1	Q10 Output voltage, high level	Open	31	40	2.4	4.5	"	"
Q2	Q11 "	"	"	39	"	"	"	"
Q3	Q12 "	"	"	38	"	"	"	"
Q4	Q13 "	"	"	37	"	"	"	"
Q5	Q20 "	"	21	30	"	"	"	"
Q6	Q21 "	"	"	29	"	"	"	"
Q7	Q22 "	"	"	28	"	"	"	"
Q8	Q23 "	"	"	27	"	"	"	"

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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load the output under test with a load as specified on figure 16.

* FIGURE 20. Output voltage (low-level and high-level) test circuit for table XV.

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*TABLE XVI. Transition time (t_{TLH} and t_{THL}) test.

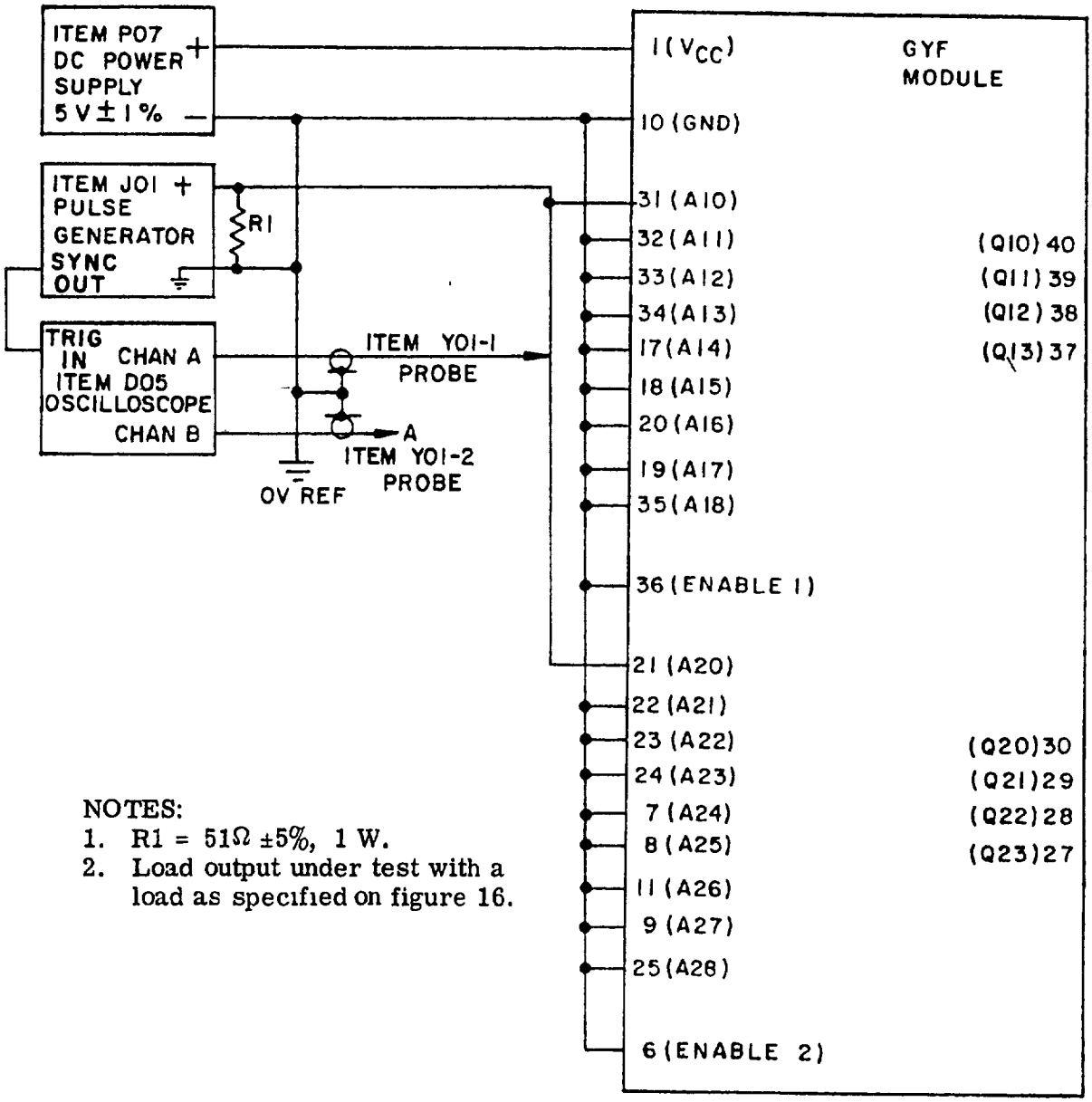
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
R1	Q10 Transition time (t_{TLH}), C_L = 50 pF	40	5	40	ns	21,22
R2	Q11 "	39	"	"	"	"
R3	Q12 "	38	"	"	"	"
R4	Q13 "	37	"	"	"	"
R5	Q20 "	30	"	"	"	"
R6	Q21 "	29	"	"	"	"
R7	Q22 "	28	"	"	"	"
R8	Q23 "	27	"	"	"	"
R9	Q10 Transition time (t_{TLH}), C_L = 150 pF	40	"	50	"	"
R10	Q11 "	39	"	"	"	"
R11	Q12 "	38	"	"	"	"
R12	Q13 "	37	"	"	"	"
R13	Q20 "	30	"	"	"	"
R14	Q21 "	29	"	"	"	"
R15	Q22 "	28	"	"	"	"
R16	Q23 "	27	"	"	"	"
S1	Q10 Transition time (t_{THL}), C_L = 50 pF	40	3	30	"	"
S2	Q11 "	39	"	"	"	"
S3	Q12 "	38	"	"	"	"
S4	Q13 "	37	"	"	"	"

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*TABLE XVI. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
S5	Q20 Transition time (t_{THL}), C_L = 50 pF	30	3	30	ns	21,22
S6	Q21 "	29	"	"	"	"
S7	Q22 "	28	"	"	"	"
S8	Q23 "	27	"	"	"	"
S9	Q10 Transition time (t_{THL}), C_L = 150 pF	40	5	40	"	"
S10	Q11 "	39	"	"	"	"
S11	Q12 "	38	"	"	"	"
S12	Q13 "	37	"	"	"	"
S13	Q20 "	30	"	"	"	"
S14	Q21 "	29	"	"	"	"
S15	Q22 "	28	"	"	"	"
S16	Q23 "	27	"	"	"	"

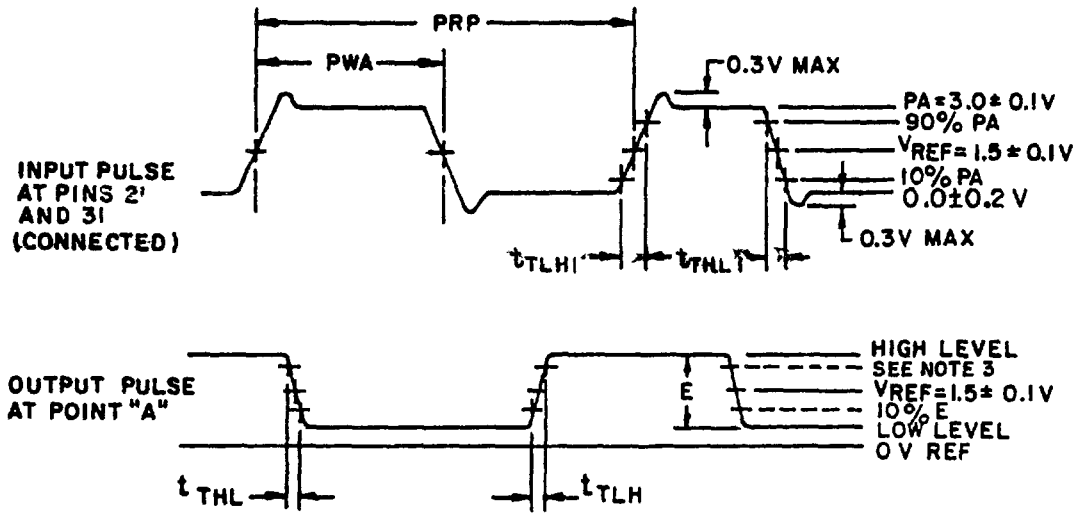
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load output under test with a load as specified on figure 16.

FIGURE 21. Transition time (t_{TLH} and t_{THL}) test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
   PWA = 200 ±50 ns.
   PRP = 400 ±50 ns.
2. Before making the measurements specified in table XVI, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.5 V.
3. For test conditions R1-R16, t_{TLH} is measured from 10% E to 80% E points.
   For test conditions S1-S16, t_{THL} is measured from 90% E to 10% E points.
4. All voltages are referenced to ground.

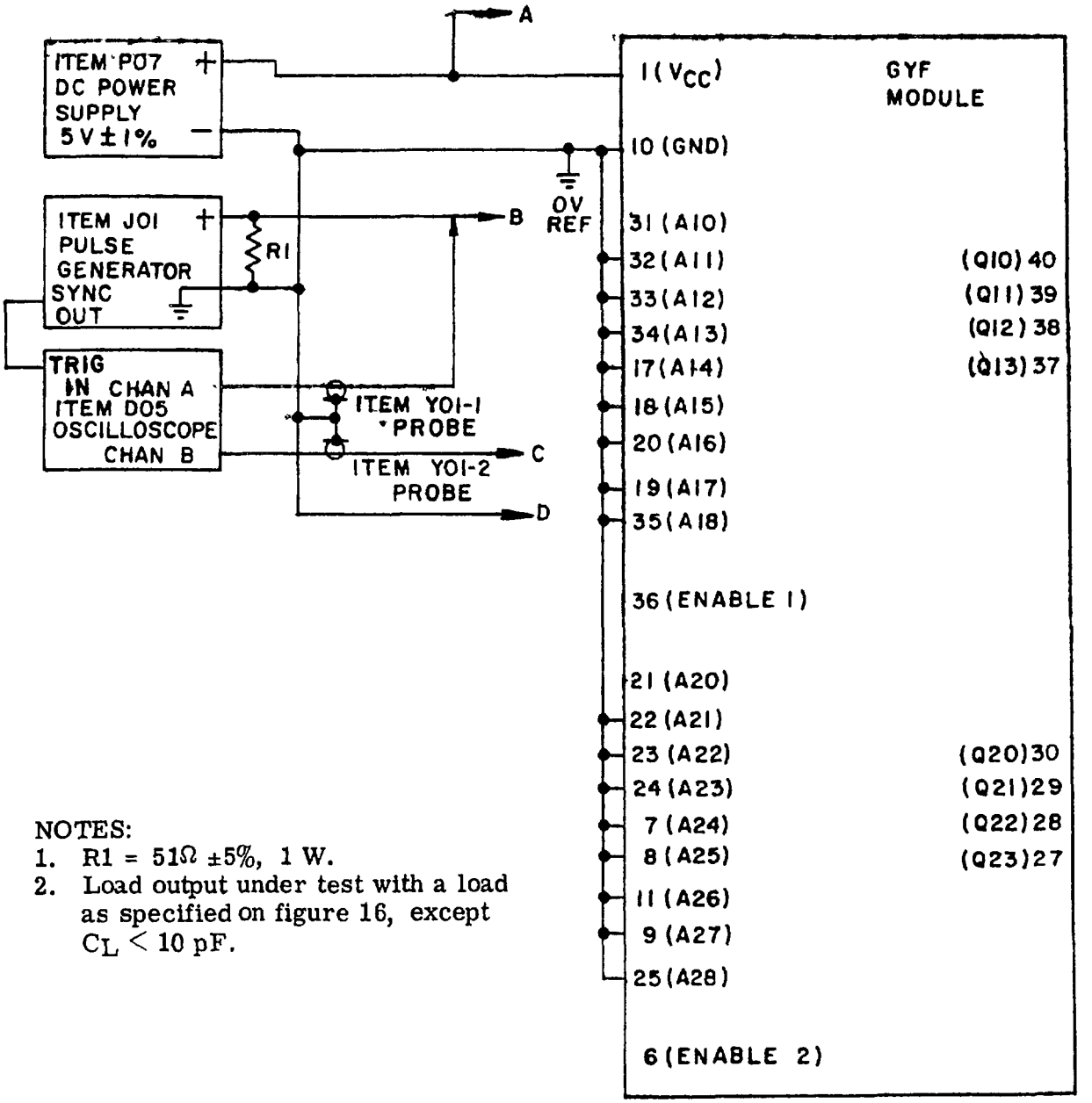
* FIGURE 22. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XVI.

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TABLE XVII. Enable recovery time (t_{ERL} and t_{ERH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	Q10 Enable recovery time (t_{ERL})	6	36	40	31,21	5	40	ns	23,24
T2	Q11 "	"	"	39	"	"	"	"	"
T3	Q12 "	"	"	38	"	"	"	"	"
T4	Q13 "	"	"	37	"	"	"	"	"
T5	Q20 "	36	6	30	"	"	"	"	"
T6	Q21 "	"	"	29	"	"	"	"	"
T7	Q22 "	"	"	28	"	"	"	"	"
T8	Q23 "	"	"	27	"	"	"	"	"
U1	Q10 Enable recovery time (t_{ERH})	6,31,21	36	40	Open	"	"	"	"
U2	Q11 "	"	"	39	"	"	"	"	"
U3	Q12 "	"	"	38	"	"	"	"	"
U4	Q13 "	"	"	37	"	"	"	"	"
U5	Q20 "	36,31,21	6	30	"	"	"	"	"
U6	Q21 "	"	"	29	"	"	"	"	"
U7	Q22 "	"	"	28	"	"	"	"	"
U8	Q23 "	"	"	27	"	"	"	"	"

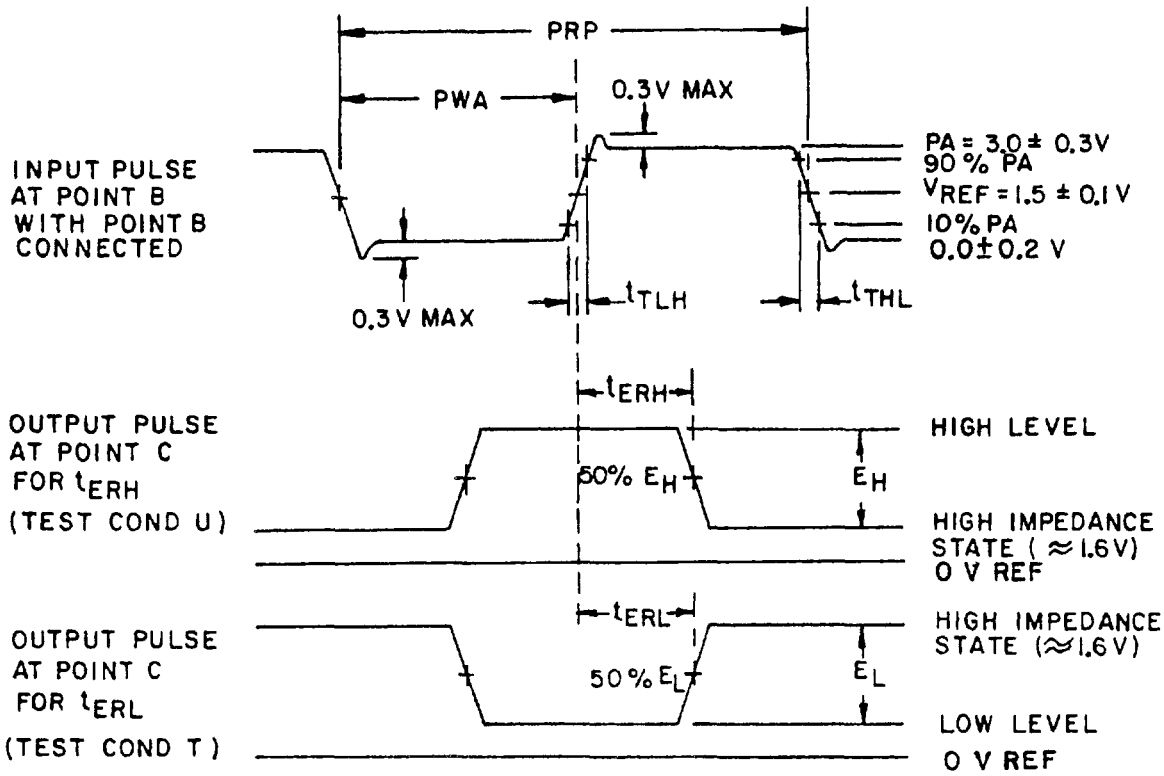
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load output under test with a load as specified on figure 16, except $C_L < 10$ pF.

FIGURE 23. Enable recovery time (t_{ERL} and t_{ERH}) test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PRP = 1, 000 ±100 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Before making the measurements specified in table XVII, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pin and between probes Y01-1 and Y01-2 and input and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

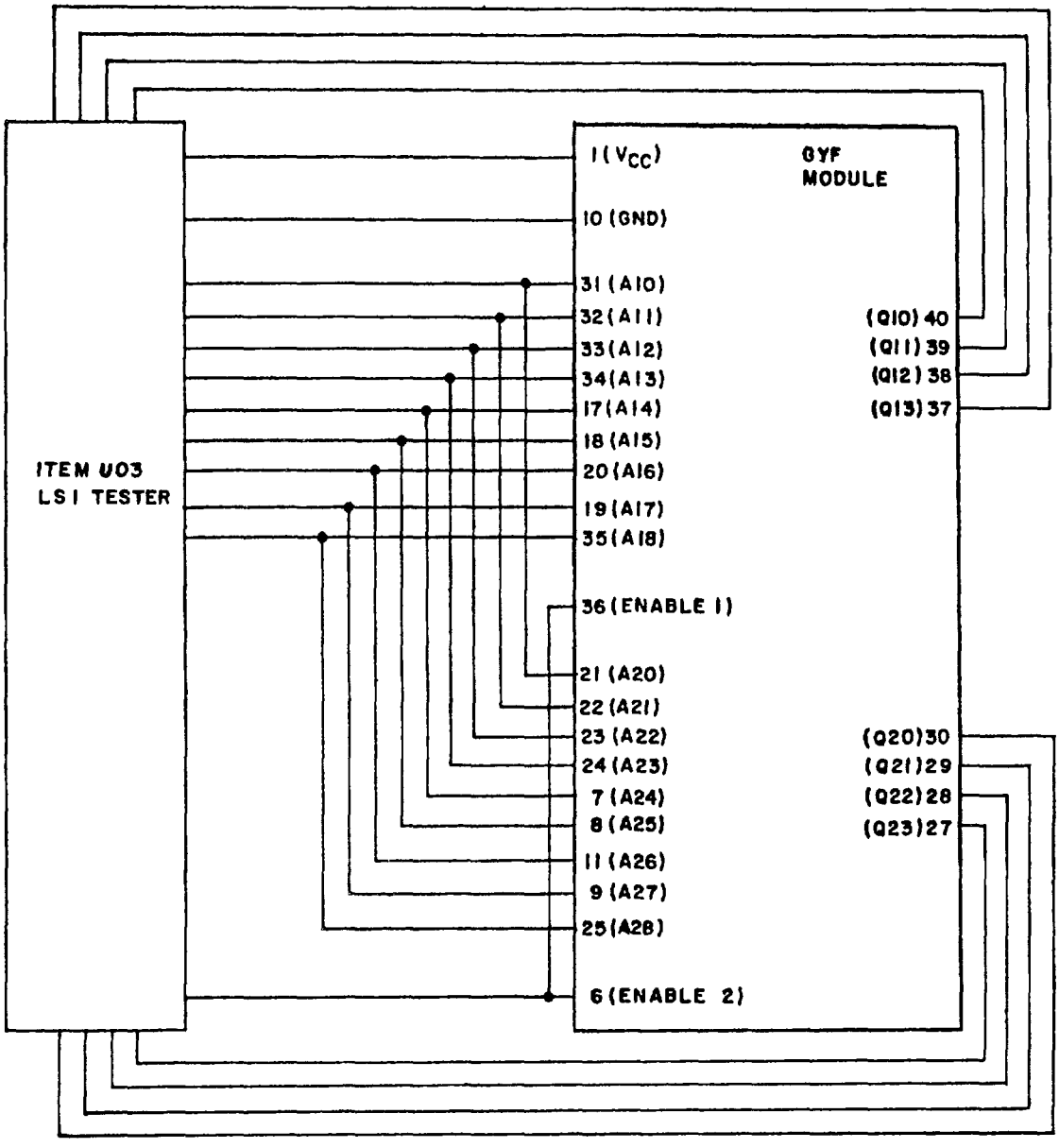
FIGURE 24. Enable recovery time (t_{ERL} and t_{ERH}) test circuit waveforms for table XVII.

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TABLE XII. Galloping read access pattern test.

Test condition	Parameter		Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
V1	Galloping read access pattern (C_L = 50 pF, V_{CC} = 4.5 V)	Module shall conform to the requirements of figures 25 and 26 for each test condition.	No failure			
V2	Galloping read access pattern (C_L = 50 pF, V_{CC} = 5.5 V)		"			
V3	Galloping read access pattern (C_L = 150 pF, V_{CC} = 4.5 V)		"			
V4	Galloping read access pattern (C_L = 150 pF, V_{CC} = 5.5 V)		"			

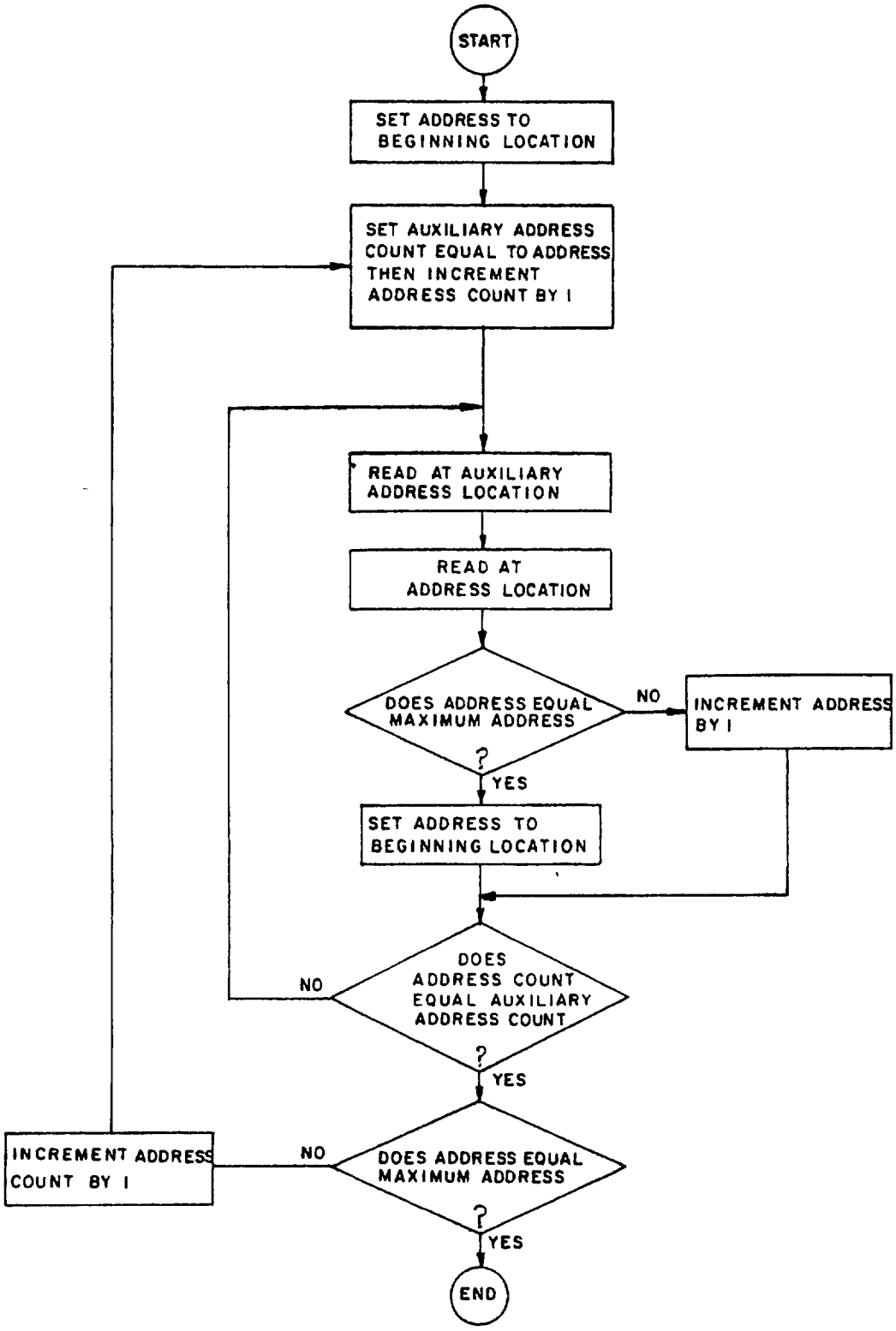
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NOTES:
1. V_{CC} = 4.5 V ±1% for test conditions V1 and V3.
2. V_{CC} = 5.5 V ±1% for test conditions V3 and V4.
3. Load output under test with a load as specified on figure 16.

FIGURE 25. Galloping read access pattern test circuit for table XVIII.

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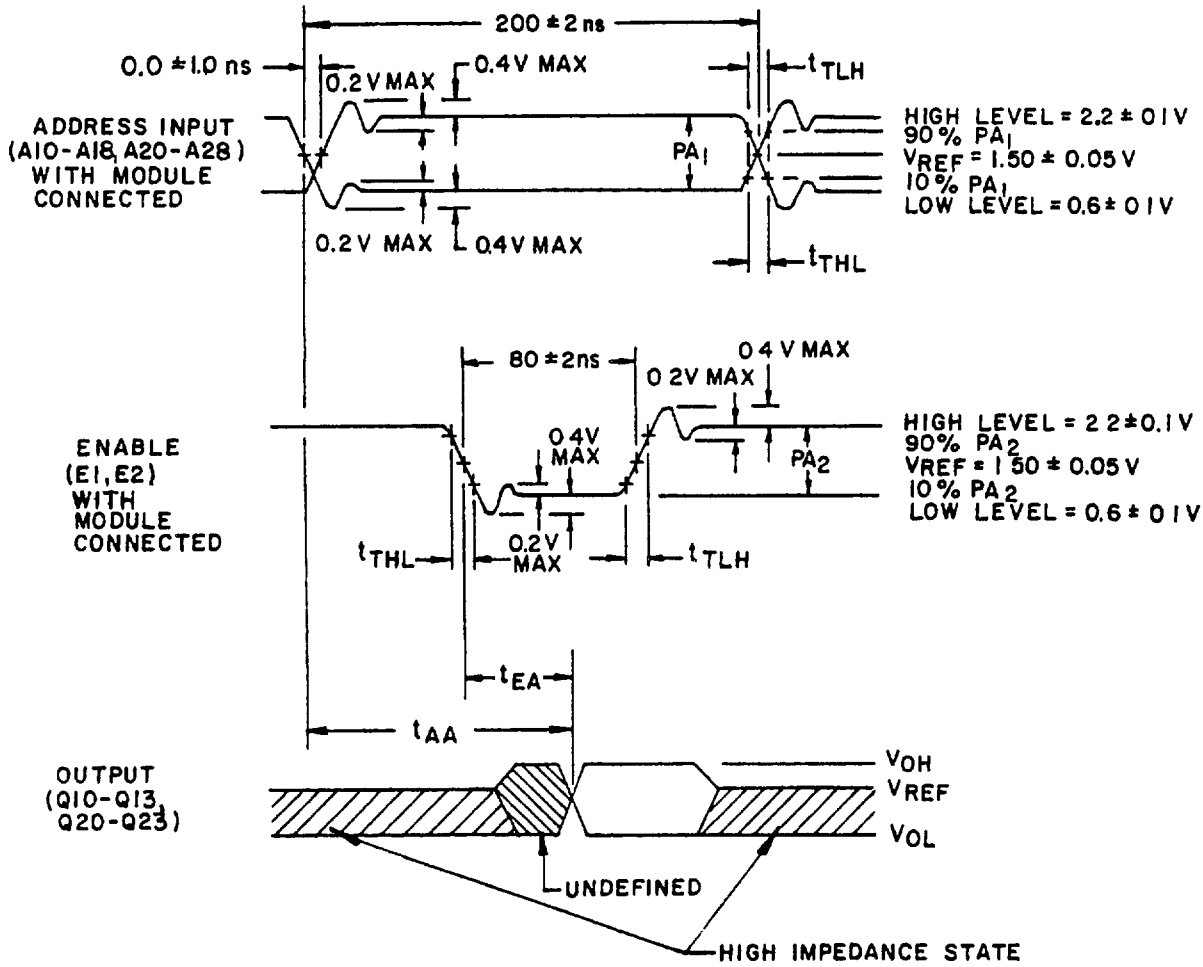


GALLOPING READ ACCESS PATTERN PROGRAM FLOW CHART

NOTE: Pattern is read and verified at each location.

FIGURE 26. Galloping read access pattern program flow chart, waveforms, and program for table XVIII.

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NOTES:
1. High impedance state = 1.6 V.
2. Address access time (50 pF load) $t_{AA1} = 80 \, ^{+0}_{-4}$ ns.
3. Address access time (150 pF load) $t_{AA2} = 90 \, ^{+0}_{-4}$ ns.
4. Enable access time (50 pF load) $t_{EA1} = 45 \, ^{+0}_{-2}$ ns.
5. Enable access time (150 pF load) $t_{EA2} = 55 \, ^{+0}_{-2}$ ns.
6. $t_{TLH} = t_{THL} = 10 \pm 4$ ns.
7. Measure $t_{EA} + t_{AA}$ for V_{OH} @ 2.4 V.
8. Measure $t_{EA} + t_{AA}$ for V_{OL} @ 0.7 V.

FIGURE 26. Galloping read access pattern program flow chart, waveforms, and program for table XVIII - Continued.

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Program for test conditions V1-V4:

1. The objective of the galloping read access pattern is to test address access time. It is necessary to test all possible address transitions. The test also verifies circuit enable time (t_{EA}) and address access time (t_{AA}).

2. A program that fulfills the galloping read access test objective would be:

(a) Read memory address zero.
(b) Read memory address one.
(c) Read memory address zero.
(d) Read memory address two.
(e) Read memory address zero.
(f) Read memory address three.

.
.
.

(g) Read memory address zero.
(h) Read memory address 511.
(i) Read memory address one.
(j) Read memory address two.
(k) Read memory address one.
(l) Read memory address three.

.
.
.

(m) Read memory address one.
(n) Read memory address 511.
(o) Read memory address one.
(p) Read memory address zero.
(q) Read memory address two.
(r) Read memory address three.
(s) Read memory address two.
(t) Read memory address four.

.
.
.

(u) Continue this pattern until all memory addresses have been tested to all other memory addresses.

FIGURE 26. Galloping read access pattern program flow chart, waveforms, and program for table XVIII - Continued.

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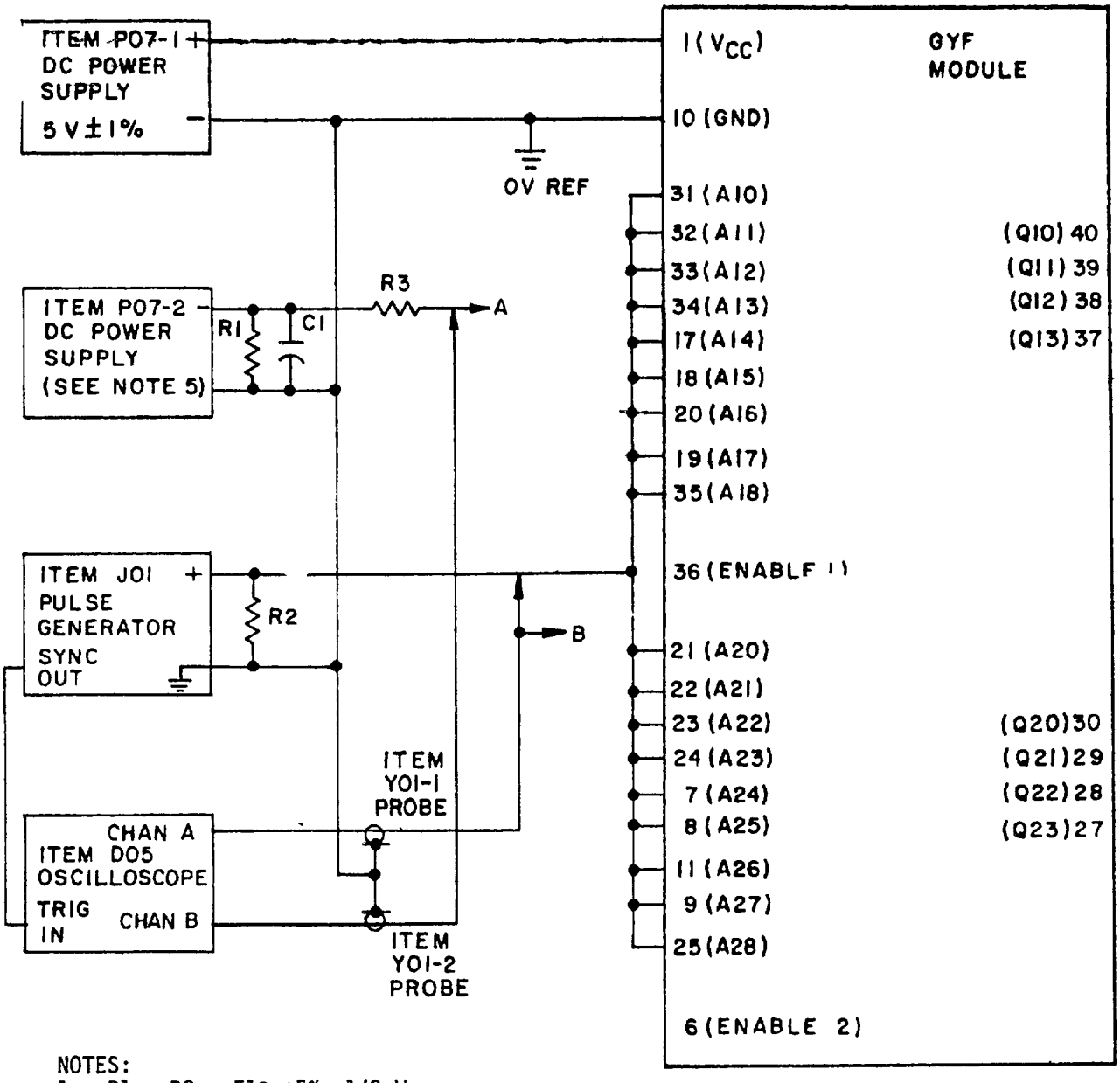
TABLE XIX. Input crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	E1 Input crosstalk, low level	36	6	0	400	mV	27,28
W2	E2 Input crosstalk, low level	6	36	"	"	"	"
X1	E1 Input crosstalk, high level	36	6	"	"	"	"
X2	E2 Input crosstalk, high level	6	36	"	"	"	"

TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient amplitude	0	400	mV	29,30

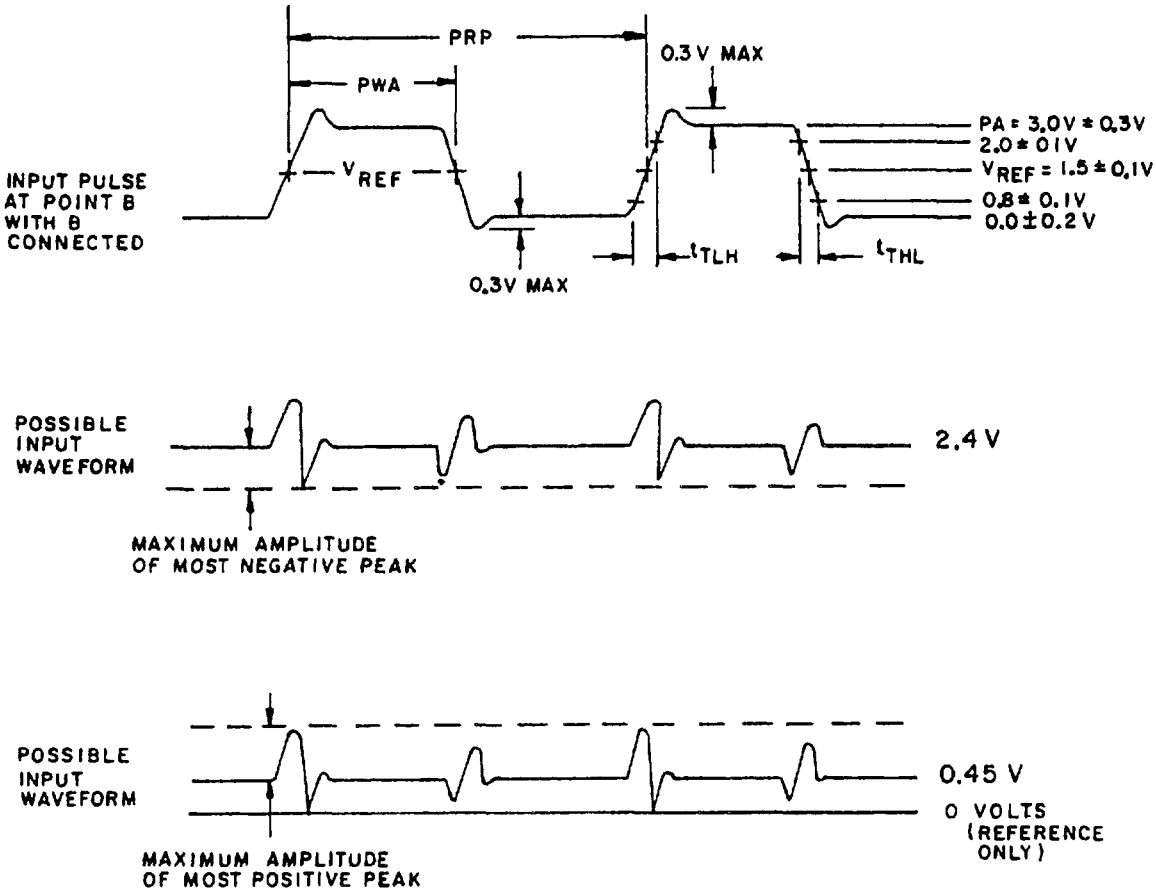
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 150Ω ±1%, 1/8 W.
3. C1 = Add sufficient capacitance to reduce item P07-2 power supply noise to < 50 mV.
4. Load all outputs with a load as specified on figure 16.
5. For test conditions W1 and W2 set item P07-2 to 0.5 V ±1%. For test conditions X1 and X2 set item P07-2 to 2.4 V ±1%.

FIGURE 27. Input crosstalk (low-level and high-level) test circuit for table XIX.

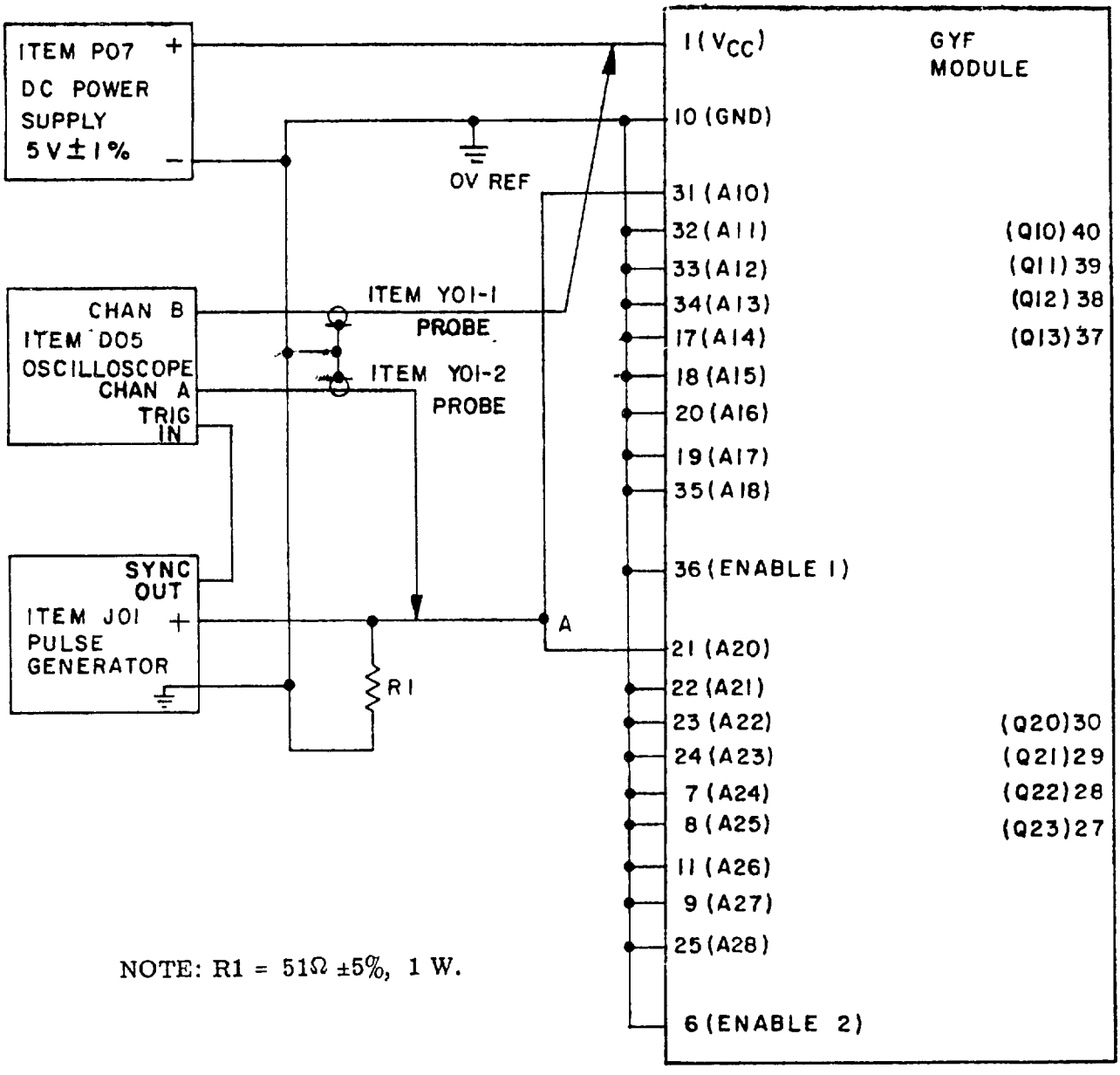
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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns verified at each input.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors shall be less than 2 inches (50.8 mm) between: resistors and input pins; probes Y01-1 and Y01-2 and input pins; load circuits and output pins; ground leads of probes Y01-1 and Y01-2 and test fixture and ground pins.

FIGURE 28. Input crosstalk (low-level and high-level) test circuit waveforms for table XIX.

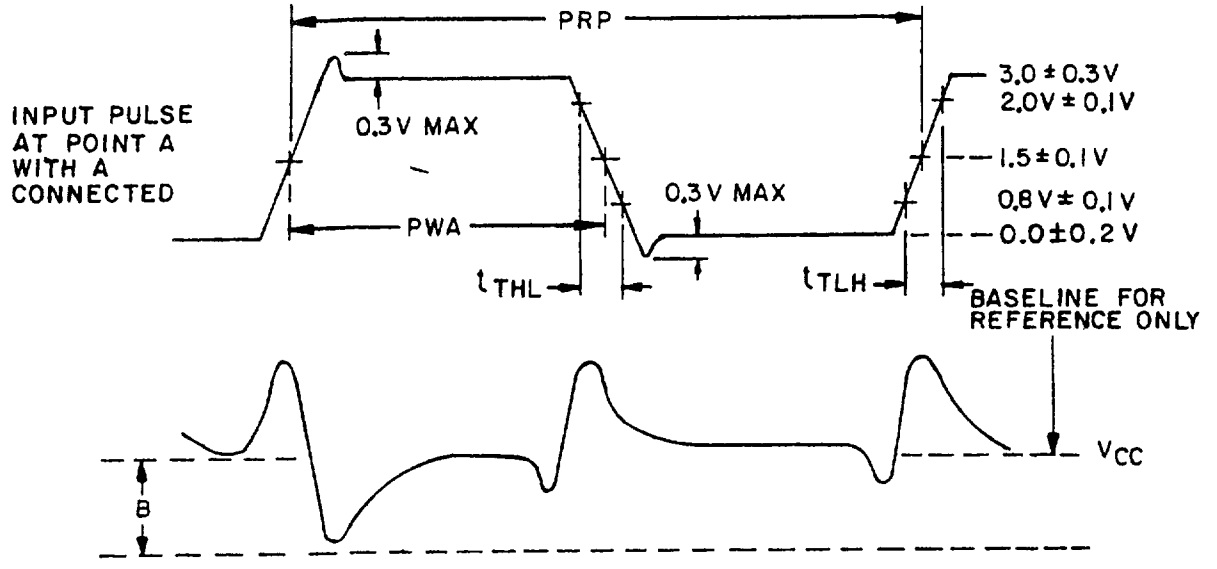
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 29. Power supply transient amplitude test circuit for table XX.

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NOTES:
1. Point "A" input parameters:
   PWA = 200 ±20 ns.
   PRP = 400 ±40 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns verified at each input.
2. B = Measurement from the baseline to the most negative excursion of the waveform.

FIGURE 30. Power supply transient amplitude test circuit waveforms for table XX.

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TABLE XXI. Input and output capacitance test.

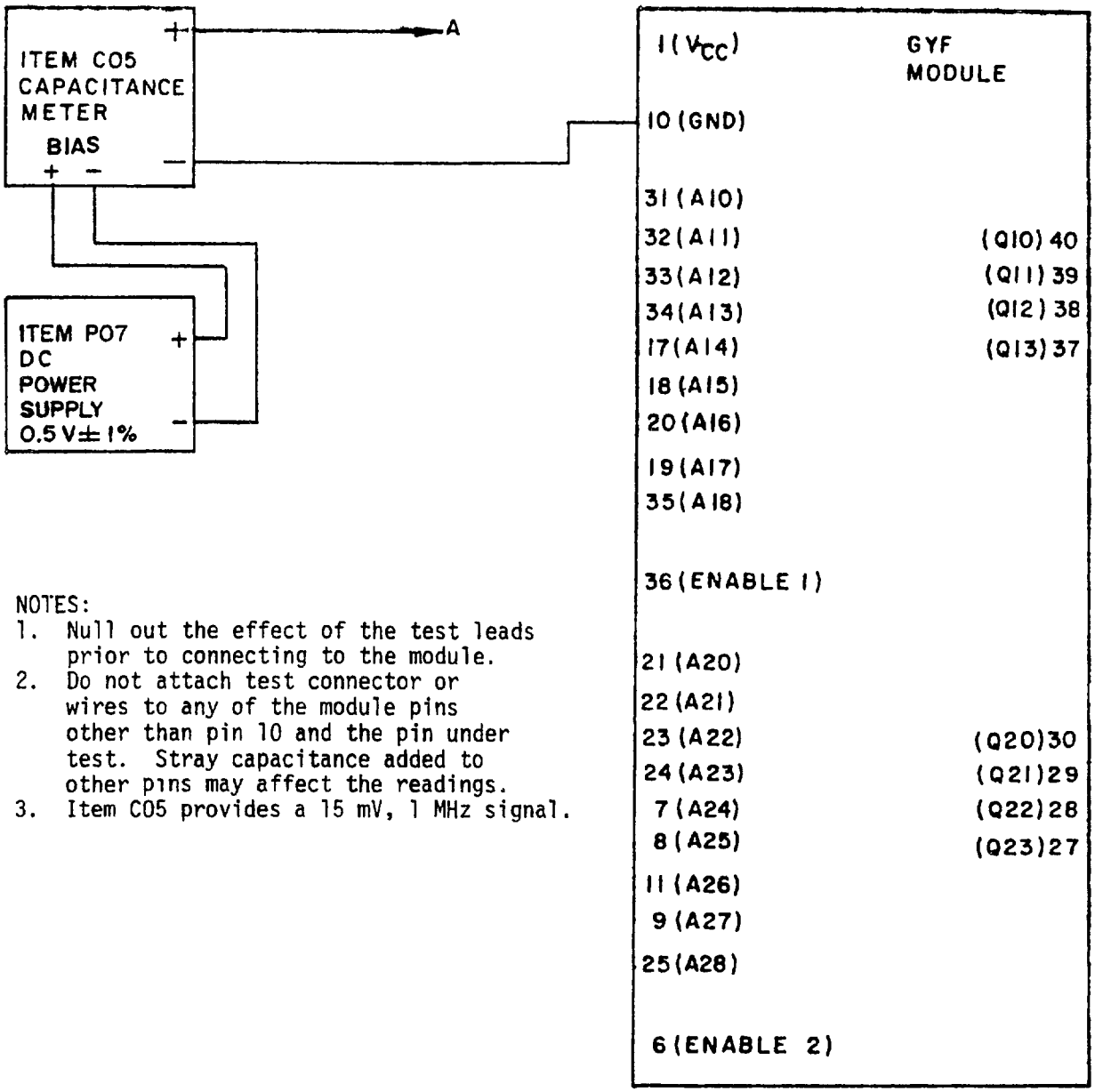
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
Z1	A10	Input capacitance	31	2	25	pF	31
Z2	A11	"	32	"	"	"	"
Z3	A12	"	33	"	"	"	"
Z4	A13	"	34	"	"	"	"
Z5	A14	"	17	"	"	"	"
Z6	A15	"	18	"	"	"	"
Z7	A16	"	20	"	"	"	"
Z8	A17	"	19	"	"	"	"
Z9	A18	"	35	"	"	"	"
Z10	A20	"	21	"	"	"	"
Z11	A21	"	22	"	"	"	"
Z12	A22	"	23	"	"	"	"
Z13	A23	"	24	"	"	"	"
Z14	A24	"	7	"	"	"	"
Z15	A25	"	8	"	"	"	"
Z16	A26	"	11	"	"	"	"
Z17	A27	"	9	"	"	"	"
Z18	A28	"	25	"	"	"	"
Z19	Enable 1	"	36	"	40	"	"
Z20	Enable 2	"	6	"	40	"	"
Z21	Q10	Output capacitance	40	"	25	"	"
Z22	Q11	Output capacitance	39	"	25	"	"

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TABLE XXI. Input and output capacitance test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
Z23	Q12	Output capacitance	38	2	25	pF	31
Z24	Q13	"	37	"	"	"	"
Z25	Q20	"	30	"	"	"	"
Z26	Q21	"	29	"	"	"	"
Z27	Q22	"	28	"	"	"	"
Z28	Q23	"	27	"	"	"	"

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C05 provides a 15 mV, 1 MHz signal.

FIGURE 31. Input and output capacitance test circuit for table XXI.

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MIL-M-28787/169B(EC)	MODULES, STANDARD ELECTRONIC MEMORY, READ ONLY, PROGRAMMABLE, DIGITAL, KEY CODE GYF

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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